

5G RAN

Random Access Control Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Expanded the application scenarios of PRACH fixed delay compensation. For details, see 5.1.9 PRACH Fixed Delay Compensation .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for 2:3 slot configuration (2_3_DSUUU) by the gNodeB in high frequency scenarios. The corresponding changes in this document are as follows: • Added information about the PUSCH SCS of 120 kHz and slot configuration of 2_3_DSUUU in Table 4-3. • Added information about the PUSCH SCS of 120 kHz and slot configuration of 2_3_DSUUU in Table 4-4. • Added a statement that PRACH coverage enhancement (high-frequency TDD) does not take effect under 2:3 slot configuration (2_3_DSUUU). For details, see 5.3.4 Cells. • Added a statement that narrow-beam probing (high-frequency TDD) cannot be enabled under 2:3 slot configuration (2_3_DSUUU). For details, see 5.3.4 Cells.	None	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the PRACH configuration indexes supported by 2:3 slot configuration (2_3_DSUUU) in high-frequency scenarios, which is detailed as follows: - Added information about PRACH configuration index 188 and slot configuration 2_3_DSUUU in Table 4-3. - Added information about PRACH configuration index 188 and slot configuration 2_3_DSUUU in Table 4-4.	None	High-frequency TDD	5900 series base stations (macro base stations)
Replaced the reserved parameter for "optimized measurement of RRC connection setup success rate based on cell radius" with a formal parameter. For details, see: 5.1.12 UE Random Access Control Based on Cell Radius 5.4 Operation and Maintenance	Replaced RSVDSWPARAM0_BIT15 of the reserved parameter NRCeLLRsvd.RsVdSwParam0 with the RRC_SETUP_SUCC_RADIUS_OPT_SW option of the NRCeLLFeatureSwitch.CounterConfigSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the PRACH configuration indexes supported by dual-period 8:2 slot configuration (8_2_DDDSUUDDDD). Specifically: - Added information about slot configuration 8_2_DDDSUUDDDD for cells with a radius greater than 14.5 km in Table 4-3. - Added information about slot configuration 8_2_DDDSUUDDDD for cells with a radius greater than 14.5 km in Table 4-4.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between doubled PRACH period and maximum preamble transmission count. For details, see 4.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)

- Editorial changes
 - Added the PRACH configuration indexes supported by dual-period 8:2 slot configuration (8_2_DDDSUUDDDD). For details, see [Table 4-3](#).
 - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

None

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Basic RA functions	None	4 Basic RA Functions
RA optimization functions	None	5 RA Optimization Functions

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Basic RA functions	Supported in both low and high frequency bands, with the following differences: • The supported RA preamble formats, PRACH configuration indexes, and the number of PRBs for the PRACH are different between high and low frequency bands. • Staggered PRACH frequency-domain resource allocation is supported only in low frequency bands.	4 Basic RA Functions

Function Name	Difference	Chapter/Section
RA optimization functions	Supported in both low and high frequency bands, with the following differences: • PRACH false alarm detection optimization is supported only in low frequency bands. • PRACH capability optimization is supported only in low frequency bands. • Narrow-beam probing is supported only in high frequency bands. • PRACH coverage enhancement is supported only in high frequency bands. • Backoff control is supported only in low frequency bands. • PRACH fixed delay compensation is supported only in low frequency bands. • UE random access control based on cell radius is supported only in low frequency bands.	5 RA Optimization Functions

3 Overview

A UE establishes a connection and achieves uplink synchronization with a gNodeB through RA for uplink transmission. During RA, the UE needs to send an RA preamble on a physical random access channel (PRACH). To ensure the RA success rate of UEs, Huawei provides the random access control function. Random access control involves the following two aspects:

- Basic RA functions: RA preamble generation and grouping, and time-frequency resource configuration for RA preambles. For details, see [4 Basic RA Functions](#).
- RA optimization functions: These functions increase the RA success rate. For details, see [5 RA Optimization Functions](#).

4 Basic RA Functions

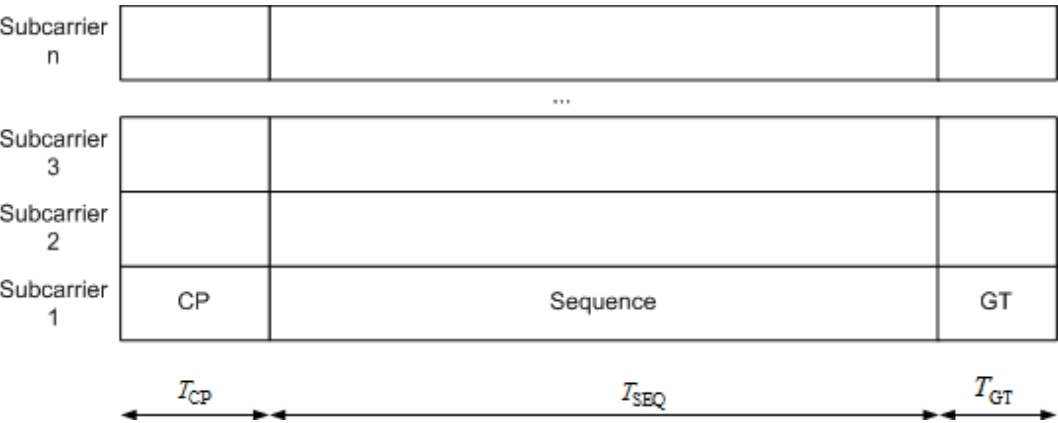
4.1 Principles

4.1.1 RA Preamble

4.1.1.1 Overview of RA Preambles

During RA, a UE needs to send an RA preamble (signal) on the PRACH. RA preambles are pulse signals with two formats: long and short. In the time domain, it consists of a T_{CP} , a T_{SEQ} , and a T_{GT} , as shown in [Figure 4-1](#). T_{CP} denotes a cyclic prefix, T_{SEQ} a preamble sequence, and T_{GT} a guard time. In the frequency domain, an RA preamble is transmitted using multiple subcarriers, specifically 839 subcarriers for a long preamble and 139 subcarriers for a short preamble.

Figure 4-1 RA preamble format



The following are the mapping relationships among format indexes, preamble lengths (L_{RA}), and PRACH SCS (Δf^{RA}) supported by the gNodeB in the current version. See [Table 4-1](#) and [Table 4-2](#). In NR TDD, the long format is supported only in low frequency bands, while the short format is supported in both low and

high frequency bands. For more information on preamble formats, see section 6.3.3.1 "Sequence generation" in 3GPP TS 38.211 (Release 15).

Table 4-1 Long preamble

Format	L_{RA}	Δf^{RA}
0	839	1.25 kHz
1	839	1.25 kHz
3	839	5.0 kHz

Table 4-2 Short preamble

Format	L_{RA}	Δf^{RA}
A3	139	$15 \times 2^{\mu}$ kHz
C2	139	$15 \times 2^{\mu}$ kHz

In this version, the preamble format and PRACH SCS are related to the duplex mode, slot configuration, PUSCH SCS, cell radius, and PRACH configuration index. Specifically:

- Duplex mode: specified by the **NRDUCell.DuplexMode** parameter. For details and data configuration, see *Cell Management*.
- Slot configuration: specified by the **NRDUCell.SlotAssignment** parameter. For details and data configuration, see *Cell Management*.
- PUSCH SCS: specified by the **NRDUCell.SubcarrierSpacing** parameter. For details and data configuration, see *Cell Management*.
- Cell radius: specified by the **NRDUCell.CellRadius** parameter. The cell radius refers to the theoretical maximum coverage distance of a cell when certain performance requirements are met. Generally, the actual coverage distance of a cell is smaller than the cell radius. For details and data configuration, see *Cell Management*.
- PRACH configuration index: specified by the **NRDUCellPrach.PrachConfigurationIndex** parameter. The value of this parameter is related to the networking scenario, system frequency band, duplex mode, slot configuration, and cell radius. For details, see *gNodeBFunction Parameter Reference*. For details about PRACH configuration index (including the mapping between PRACH configuration indexes and PRACH periods), see section 6.3.3.2 "Mapping to physical resources" in 3GPP TS 38.211 V15.7.0.

Table 4-3 lists the relationship between the preamble formats, PRACH SCS, and related parameters.

Table 4-3 Preamble formats and PRACH SCS used by the gNodeB

Duplex Mode	Slot Configuration	PUSCH SCS	Cell Radius	PRACH Configuration Index	Preamble Format	PRACH SCS
CELL_TDD	4_1_DDDSU	30 kHz	≤ 9.65 km	200, 202, 210, 65535	C2	15 kHz
CELL_TDD	8_2_DDDDDDD SUU or 7_3_DDDSUDD SUU	30 kHz	> 14.5 km and ≤ 60 km	0 to 7, 12, 17, 65535	0	1.25 kHz
CELL_TDD	8_2_DDDDDDD SUU or 7_3_DDDSUDD SUU	30 kHz	> 60 km and ≤ 90 km	43 to 47, 52, 57, 65535	3	5 kHz
CELL_TDD	8_2_DDDDDDD SUU or 7_3_DDDSUDD SUU	30 kHz	≤ 14.5 km	0 to 7, 12, 17, 65535	0	1.25 kHz
CELL_TDD	8_2_DDDDDDD SUU or 7_3_DDDSUDD SUU	30 kHz	≤ 9.65 km	200, 202, 210	C2	15 kHz
CELL_TDD	8_2_DDDSUUD DDD	30 kHz	≤ 9.65 km	200, 202, 210	C2	15 kHz
CELL_TDD	8_2_DDDSUUD DDD	30 kHz	≤ 60 km	9, 14, 19, 65535	0	1.25 kHz
CELL_TDD	8_2_DDDSUUD DDD	30 kHz	> 60 km and ≤ 90 km	49, 54, 59, 65535	3	5 kHz
CELL_TDD	4_1_DDDSU	120 kHz	≤ 2.41 km	188, 65535	C2	60 kHz
CELL_TDD	4_1_DDDSU	120 kHz	≤ 5 km	192	C2	60 kHz
CELL_TDD	4_1_DDDSU	120 kHz	≤ 1.16 km	176 ^a , 180, 189	C2	120 kHz
CELL_TDD	4_1_DDDSU	120 kHz	≤ 0.96 km	74	A3	60 kHz
CELL_TDD	2_3_DSUUU	120 kHz	≤ 2.41 km	188, 65535	C2	60 kHz
CELL_TDD	2_3_DSUUU	120 kHz	≤ 1.16 km	186, 65535	C2	120 kHz
CELL_SUL	N/A	15 kHz	≤ 14.5 km	0 to 21, 65535	0	1.25 kHz

a: The PRACH configuration index can be set to 176 only when the slot configuration is 4:1, the SCS is 120 kHz, and the **NRDUCellTrpMmwavBeam.CoverageScenario** parameter is not set to **DEFAULT** or **SCENARIO_102**.

 NOTE

In NR TDD, it is recommended that the **NRDUCellPrach.PrachConfigurationIndex** parameter be set to **202** or **206** for low-frequency cells in certain scenarios. For details, see [4.3.4 Cells](#).

4.1.1.2 Preamble Sequence Generation

Preamble sequences are generated from cyclic shifts of Zadoff-Chu (ZC) root sequences and are also called physical root sequence indexes or physical root sequences.

- The start logical ZC root sequence index is specified by the **NRDUCell.LogicalRootSequenceIndex** parameter. For a long preamble, the parameter value cyclically ranges from 0 to 837. For a short preamble, the parameter value cyclically ranges from 0 to 137. This parameter indicates the index of the start logical root sequence used to generate the cell preamble sequence. Each logical root sequence corresponds to one physical root sequence. For details, see 3GPP TS 38.211.

The following are examples:

- For a long preamble, logical index 837 is followed by index 0.
- For a short preamble, logical index 137 is followed by index 0.
- The gNodeB determines the number of cyclic shifts based on the cell radius, preamble format, and PUSCH SCS.

A maximum of 64 preambles can be generated for each cell. If fewer than 64 preamble sequences are generated from a single ZC root sequence, subsequent ZC root sequences are used until 64 preambles are generated. For more information on preamble sequence generation, see section 6.3.3.1 "Sequence generation" in 3GPP TS 38.211 V2.0.0. The logical ZC sequence index and the number of cyclic shifts are indicated by the *RACH-ConfigCommon* IE.

 NOTE

In SA networking, the *RACH-ConfigCommon* IE is carried in SIB1. In NSA networking, the *RACH-ConfigCommon* IE is carried in the RRCConnectionReconfiguration message.

For details about the logical root sequence index (specified by **NRDUCell.LogicalRootSequenceIndex**) and related data configuration, see *Cell Management*.

4.1.1.3 Preamble Sequence Grouping

A cell has 64 preamble sequences, which are used during requests for system information and RA.

- If on-demand system information transmission is enabled for a cell, some preamble sequences are used during requests for system information, and other preamble sequences are used during RA.
- If on-demand system information transmission is not enabled for a cell, all the preamble sequences are used during RA.

 NOTE

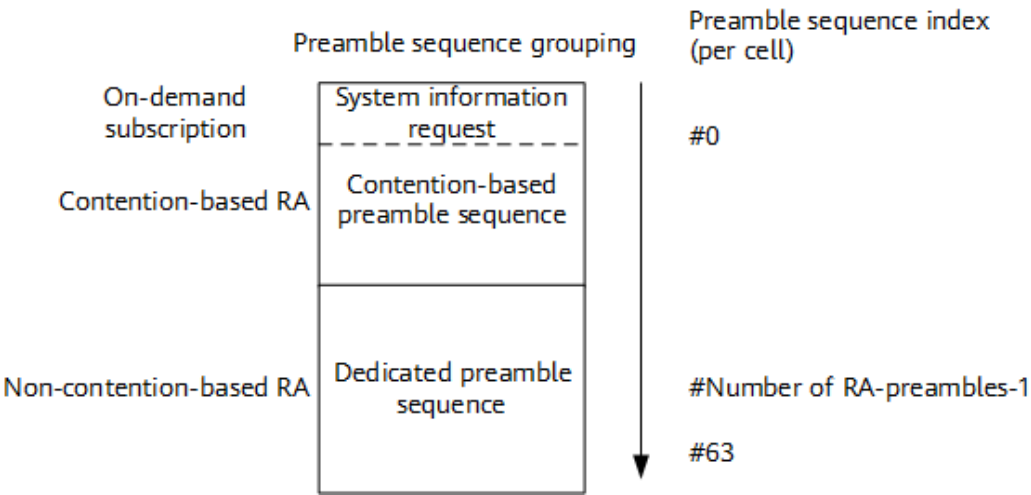
For details about the system information transmission policies of cells, see *5G Networking and Signaling*.

Based on RA contention, the preamble sequences for RA are classified into contention-based preamble sequences and dedicated preamble sequences, as [Figure 4-2](#) illustrates. The proportion of contention-based preamble sequences to the total contention-based and dedicated preamble sequences can be configured by the **NRDUCellPrach.CbraPreamblePct** parameter.

- If this parameter is set to a value other than **255**, the actual proportion of contention-based preamble sequences is the value of this parameter.
- If this parameter is set to **255**, the actual proportion of contention-based preamble sequences is 82% for low-speed cells (for which **NRDUCell.HighSpeedFlag** is set to **LOW_SPEED**) or 30% for high-speed cells (for which **NRDUCell.HighSpeedFlag** is set to **HIGH_SPEED**).

The number of contention-based preambles varies with RA scenarios and is indicated by **totalNumberOfRA-Preambles** in the *RACH-ConfigCommon* IE. For more information, see section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V2.0.0.

Figure 4-2 Preamble sequence grouping



4.1.2 Time-Frequency Resources for RA

During RA, a UE needs to send an RA preamble on a specific time-frequency resource, which is PRACH.

Time-Domain Position

The time-domain position of the PRACH refers to the frame number and slot number of a preamble, which are determined by the PRACH configuration index. For more information, see section 6.3.3.2 "Mapping to physical resources" in 3GPP TS 38.211 V2.0.0. The PRACH configuration index used by the gNodeB is specified by the **NRDUCellPrach.PrachConfigurationIndex** parameter.

- If this parameter is not set to **65535**, the PRACH configuration index used by the gNodeB is the value of this parameter. For details about the configuration, see [Table 4-3](#).
- If this parameter is set to **65535**, the PRACH configuration index used by the gNodeB is automatically generated by the gNodeB based on the system frequency band, duplex mode (**NRDUCell.DuplexMode**), slot configuration (**NRDUCell.SlotAssignment**), PUSCH SCS (**NRDUCell.SubcarrierSpacing**), and cell radius (**NRDUCell.CellRadius**). [Table 4-4](#) lists the PRACH configuration indexes that are automatically generated by the gNodeB based on the parameters in the current version.

Table 4-4 PRACH configuration indexes automatically generated by the gNodeB based on parameters (when **NRDUCellPrach.PrachConfigurationIndex** is set to **65535**)

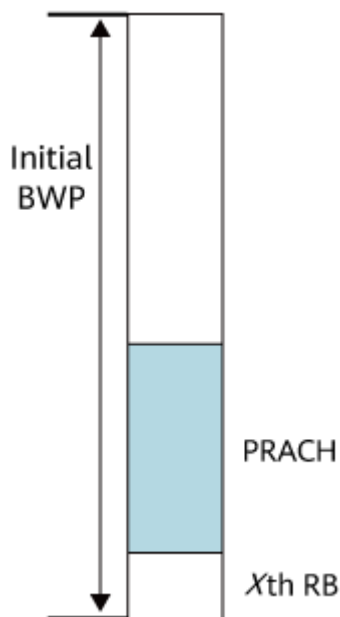
Duplex Mode	Slot Configuration	PUSCH SCS	Cell Radius	PRACH Configuration Index
CELL_TDD	4_1_DDDSU	30 kHz	≤ 9.65 km	210
CELL_TDD	8_2_DDDDDDDSUU or 7_3_DDDSUDDSUU	30 kHz	> 14.5 km and ≤ 60 km	17
CELL_TDD	8_2_DDDDDDDSUU or 7_3_DDDSUDDSUU	30 kHz	> 60 km and ≤ 90 km	57
CELL_TDD	8_2_DDDDDDDSUU or 7_3_DDDSUDDSUU	30 kHz	≤ 14.5 km	17
CELL_TDD	8_2_DDDSUUDDDD	30 kHz	≤ 60 km	19
CELL_TDD	8_2_DDDSUUDDDD	30 kHz	> 60 km and ≤ 90 km	59
CELL_TDD	4_1_DDDSU	120 kHz	≤ 2.41 km	188
CELL_TDD	2_3_DSUUU	120 kHz	≤ 1.16 km	186
CELL_TDD	2_3_DSUUU	120 kHz	> 1.16 km and ≤ 2.41 km	188
CELL_SUL	N/A	15 kHz	≤ 14.5 km	• SA networking: 19 • NSA networking: 21 • NSA and SA hybrid networking: 21

Frequency-Domain Position

The start frequency-domain position of the PRACH is specified by the **NRDUCell.PrachFreqStartPosition** parameter.

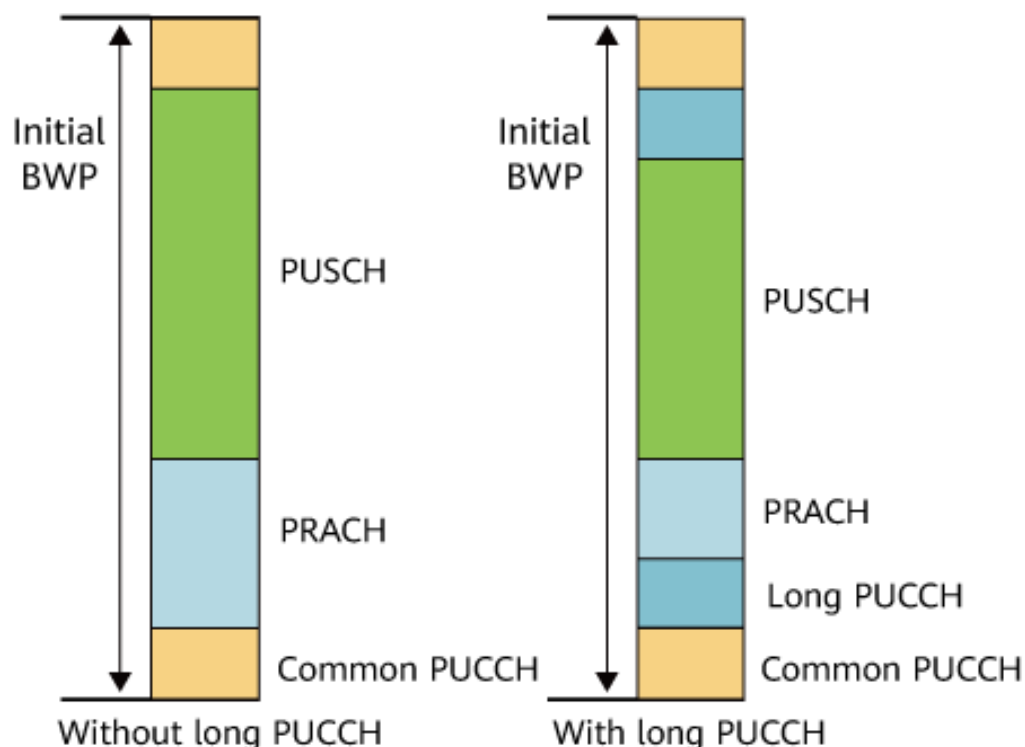
- If this parameter is not set to **65535**, the start frequency-domain position of the PRACH is the X th RB of the initial BWP where X is the value of this parameter.

Figure 4-3 Frequency-domain position of the PRACH (with a parameter value other than **65535**)



- If this parameter is set to **65535**, the PRACH is located at the lower end of the initial BWP. If the long PUCCH is configured in the initial BWP, the PRACH is adjacent to the long PUCCH. If the long PUCCH is not configured in the initial BWP, the PRACH is adjacent to the common PUCCH, as shown in [Figure 4-4](#). For details about the long PUCCH, see *Channel Management*. The common PUCCH is indicated by *pucch-ResourceCommon*. For details about the common PUCCH, see section 9.2.1 "PUCCH Resource Sets" in 3GPP TS 38.213 V15.5.0.

Figure 4-4 Frequency-domain position of the PRACH (with a parameter value of 65535)



In low-frequency TDD, if the preamble format used by the gNodeB is C2 (described in [Table 4-3](#)), PRACH root sequence conflicts cannot be avoided by replanning the logical root sequence indexes of neighboring cells. To solve this problem, staggered PRACH frequency-domain resource allocation is introduced. This function is controlled by the **PRACH_FREQ_POS_STAGGER_SW** option of the **NRDUCellPrach.PrachAlgoExtSwitch** parameter.

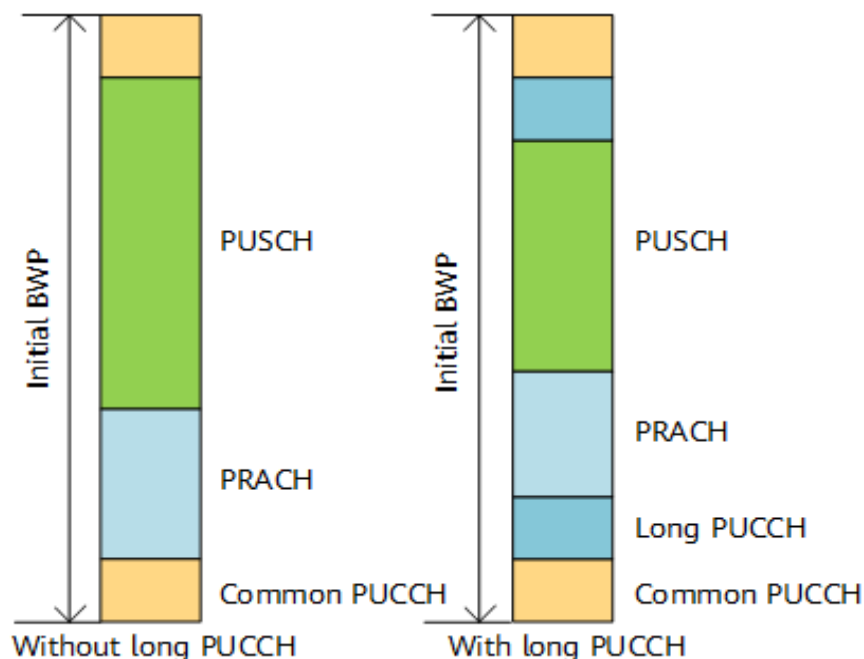
- When the **NRDUCellPrach.PrachConfigurationIndex** parameter is set to **200**, **202**, or **210**, this function can be enabled in low-frequency TDD cells.
- When the **NRDUCellPrach.PrachConfigurationIndex** parameter is set to **65535**, this function can be enabled only in the following cells:
 - Low-frequency TDD high-speed cells (**NRDUCell.HighSpeedFlag** set to **HIGH_SPEED**)
 - Low-frequency TDD low-speed cells (**NRDUCell.HighSpeedFlag** set to **LOW_SPEED**) with a slot configuration of 4_1_DDDSU (**NRDUCell.SlotAssignment** set to **4_1_DDDSU**)
- When the **NRDUCellPrach.PrachConfigurationIndex** parameter is set to any other value, this function cannot be enabled.

When staggered PRACH frequency-domain resource allocation is enabled and the **NRDUCell.PrachFreqStartPosition** parameter is set to **65535**, the PRACH

is located at the lower end of the initial BWP for cells with PCI modulo 2 being 0 and at the higher end of the initial BWP for cells with PCI modulo 2 being 1. Specifically:

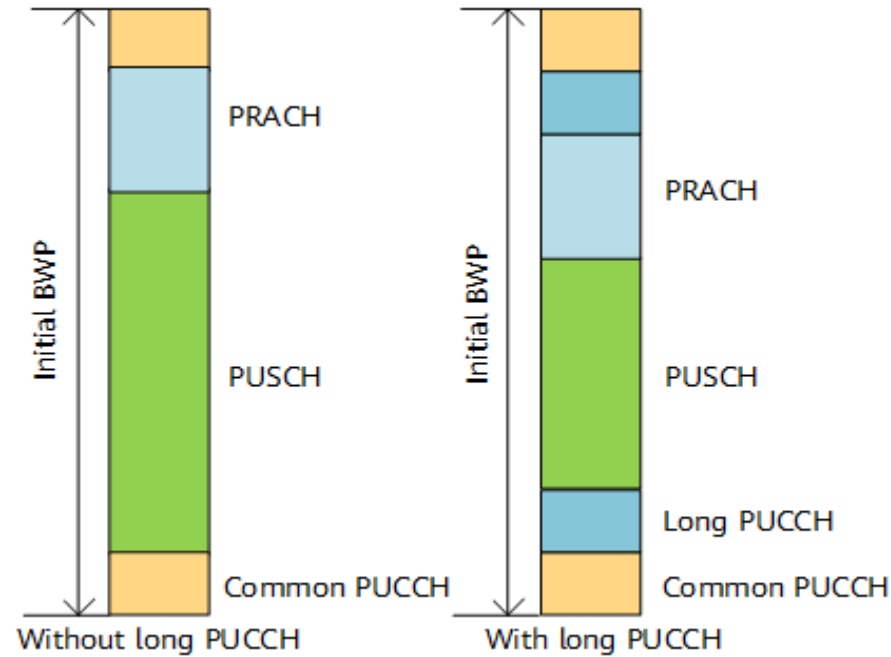
- For cells whose PCI modulo 2 is 0, the PRACH is located at the lower end of the initial BWP. (See [Figure 4-5](#).)
- For cells whose PCI modulo 2 is 1, the PRACH is located at the higher end of the initial BWP. (See [Figure 4-6](#).)

Figure 4-5 Frequency-domain position of the PRACH (with a parameter value of 65535 for cells with PCI modulo 2 being 0)



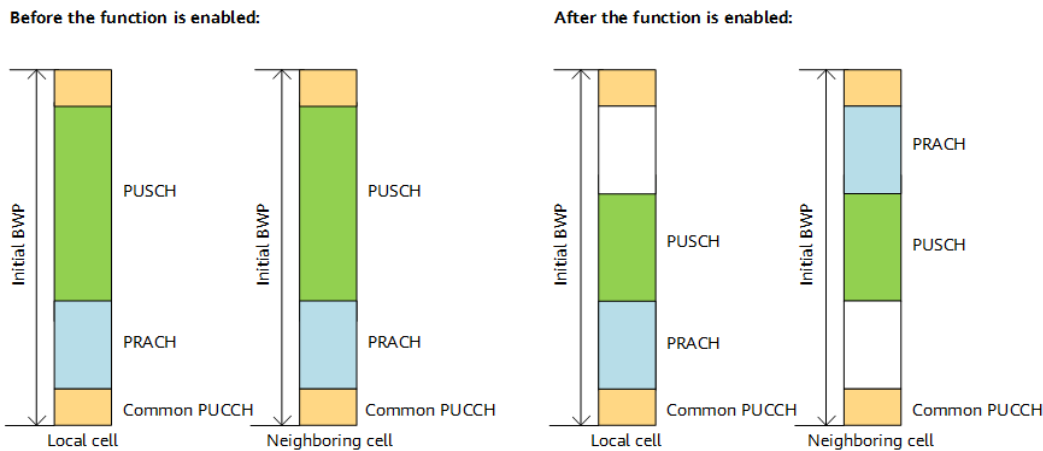
If the long PUCCH is configured in the initial BWP, the PRACH is adjacent to the long PUCCH. If the long PUCCH is not configured in the initial BWP, the PRACH is adjacent to the common PUCCH.

Figure 4-6 Frequency-domain position of the PRACH (with a parameter value of 65535 for cells with PCI modulo 2 being 1)



In addition, after staggered PRACH frequency-domain resource allocation is enabled, if there are no UEs with heavy uplink traffic in the cell, the PUSCH frequency-domain resource range excludes the PRBs used by the PRACH in neighboring cells. If there are UEs with heavy uplink traffic in the cell, the PUSCH frequency-domain resource range will remain unchanged. For details about the PUSCH frequency-domain resource range, see *Uplink Scheduling*. [Figure 4-7](#) shows the PUSCH frequency-domain resource ranges before and after staggered PRACH frequency-domain resource allocation is enabled. This case assumes that the long PUCCH is not configured in the initial BWP, there are no UEs with heavy uplink traffic in the cell, the PCI modulo 2 value of the local cell is 0, and that of the neighboring cell is 1.

Figure 4-7 PUSCH frequency-domain resource range



The PRACH occupies multiple PRBs in the frequency domain. The number of PRBs depends on the preamble length (L_{RA}), PRACH SCS (Δf^{RA}), and PUSCH SCS

(Δf). **Table 4-5** lists the number of PRBs supported by Huawei gNodeBs. For more information on all the quantities of PRBs defined in 3GPP specifications, see section 6.3.3.2 "Mapping to physical resources" in 3GPP TS 38.211 (Release 15).

Table 4-5 Number of PRBs for the PRACH

L_{RA}	Δf^{RA} for PRACH	Δf for PUSCH	Number of PRBs for the PRACH
839	1.25 kHz	15 kHz	6
839	1.25 kHz	30 kHz	3
839	5 kHz	30 kHz	12
139	15 kHz	30 kHz	6
139	60 kHz	120 kHz	6
139	120 kHz	120 kHz	12

4.1.3 RA Procedures

4.1.3.1 RA Procedures in SA Networking

After the cell search is complete, a UE achieves downlink synchronization with the cell and can receive downlink data. However, the UE has not achieved uplink synchronization with the cell yet. The UE establishes a connection with the cell and achieves uplink synchronization through RA to perform uplink transmission. The UE initiates RA using specific PRACH time-frequency resources. When the UE initiates RA, an RA preamble is transmitted. The RA preamble is used to inform the gNodeB of an RA request, allowing the gNodeB to estimate the transmission delay between the gNodeB and the UE.

RA Mechanism

There are two RA mechanisms:

- Contention-based RA
Preambles are randomly selected by UEs and such preambles are random. If preambles selected by different UEs conflict, the gNodeB uses a contention resolution mechanism to handle access requests. The RA result is random and not all RA procedures succeed.
- Non-contention-based RA
Preambles are allocated by the gNodeB to UEs and such preambles are dedicated. Therefore, there is no preamble conflict. When dedicated preambles are insufficient, the gNodeB instructs UEs to initiate contention-based RA.

The contention-based RA procedure slightly differs from the non-contention-based RA procedure. For details about RA procedures, see section 5.1 "Random Access procedure" in 3GPP TS 38.321 V15.5.0.

The RA mechanism selected varies with the triggering scenario. For details about RA triggering scenarios and mechanisms, see [Table 4-6](#).

Table 4-6 RA triggering scenarios and mechanisms

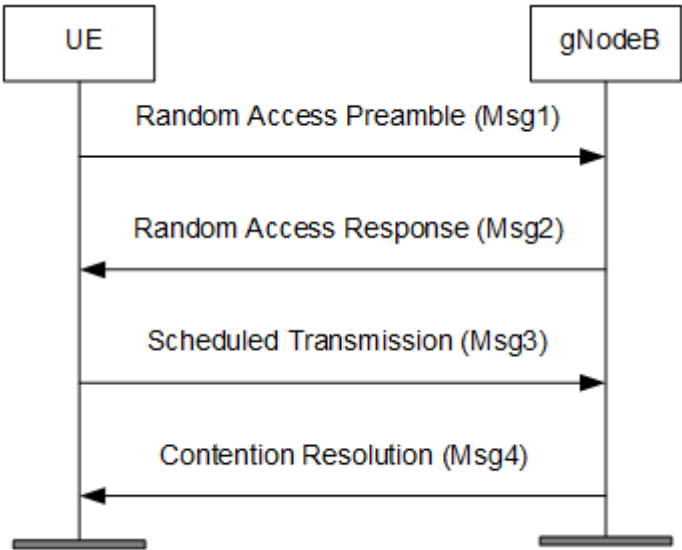
Triggering Scenario	Scenario Description	Mechanism
Initial RRC connection setup	When a UE needs to transit from RRC_IDLE to RRC_CONNECTED, the UE initiates RA to establish an RRC connection. - When the UE in the RRC_IDLE state proactively transits to the RRC_CONNECTED state, the UE initiates RA. - When the UE is in the RRC_IDLE state and the gNodeB needs the UE to transit to the RRC_CONNECTED state, the gNodeB sends a paging message to instruct the UE to initiate RA.	Contention-based RA
RRC connection reestablishment	After detecting a radio link failure, a UE initiates RA to reestablish an RRC connection.	Contention-based RA
Handover	During a handover, a UE initiates RA in the target cell.	Non-contention-based RA is the first choice. Contention-based RA is used when dedicated preambles of the gNodeB are used up.
Downlink data arrival	When a gNodeB needs to send downlink data to an RRC_CONNECTED UE in an uplink out-of-synchronization state, the gNodeB instructs the UE to initiate RA.	Non-contention-based RA is preferentially used if the UE reports the SSB index. In other scenarios, contention-based RA is used.
Uplink data transmission	When an RRC_CONNECTED UE in an uplink out-of-synchronization state needs to send uplink data to a gNodeB, the UE initiates RA.	Contention-based RA

Triggering Scenario	Scenario Description	Mechanism
Transition from RRC_INACTIVE to RRC_CONNECTED	When a UE needs to transit from RRC_INACTIVE to RRC_CONNECTED, the UE initiates RA. • When the UE in the RRC_INACTIVE state proactively transits to the RRC_CONNECTED state, the UE initiates RA. • When the UE is in the RRC_INACTIVE state and the gNodeB needs the UE to transit to the RRC_CONNECTED state, the gNodeB sends a paging message to instruct the UE to initiate RA.	Contention-based RA
Msg1-based OSI request	When a type of OSI is delivered in on-demand delivery mode, a UE requests this type of OSI through Msg1.	Non-contention-based RA
Msg3-based OSI request	When a type of OSI is delivered in on-demand delivery mode, a UE requests this type of OSI through RRCSystemInfoRequest.	Contention-based RA

4.1.3.1.1 Contention-based RA Procedure

Figure 4-8 shows the contention-based RA procedure.

Figure 4-8 Contention-based RA procedure

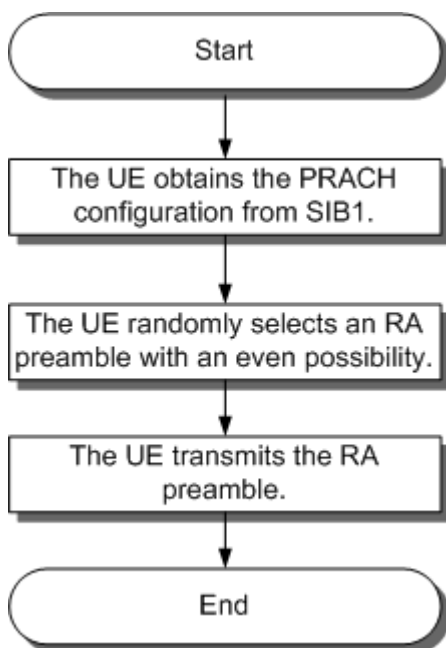


The contention-based RA procedure is as follows:

1. The UE sends an RA preamble.

In initial access scenarios, the UE sends a Random Access Preamble message (Msg1). [Figure 4-9](#) shows the procedure for sending an RA preamble. In handover scenarios, the *MobilityControlInfo* IE sent by the source gNodeB carries the RA preamble allocated to the UE.

Figure 4-9 RA preamble transmission procedure



- a. The UE obtains the PRACH configuration from SIB1.

The UE obtains the time-frequency domain position for transmitting an RA preamble based on the PRACH configuration.

According to 3GPP TS 38.211 V16.7.0, the time-domain position of a preamble is indicated by the higher-layer parameter **prach-ConfigurationIndex** or **msgA-PRACH-ConfigurationIndex**. Its time-domain information such as the preamble format, frame number, subframe number, starting symbol, and number of consecutive symbols is determined by checking Tables 6.3.3.2-2 to 6.3.3.2-4 in 3GPP TS 38.211 V16.7.0.

- b. The UE randomly selects an RA preamble with an even possibility.

Each SSB corresponds to a unique preamble index. Therefore, before selecting an RA preamble, the UE first selects an SSB, and then determines the RA preamble based on the SSB. For details, see section 5.1.2 "Random Access Resource selection" in 3GPP TS 38.321 V15.5.0. The UE compares the RSRP values of all SSBs in the cell with the RSRP threshold and selects an SSB whose RSRP value is greater than the threshold. The RSRP threshold is indicated by the *rsrp-ThresholdSSB* IE in 3GPP TS 38.331 V15.5.0 and specified by the **NRDUCellPrach.RsrpThldForSsbSelection** parameter. If no SSB meets the requirement, the UE selects an SSB randomly. For details, see section 5.1.2 "Random Access Resource selection" in 3GPP TS 38.321 V15.5.0.

- c. The UE sends the RA preamble.
- 2. The gNodeB sends an RA response.

After receiving the RA preamble from the UE, the gNodeB obtains the timing alignment information. The gNodeB sends an RA response to the UE over the PDSCH through Msg2. One PDSCH can carry RA responses to multiple UEs at the same time. The specific information carried in the RA response includes: RA-preamble identifier, timing alignment information, UL grant, and temporary C-RNTI.

After the UE sends the RA preamble, it monitors the PDCCH within the RA response window until an RA response is received over the PDSCH.

- If the UE receives a response containing an RA-preamble identifier which is the same as the identifier contained in the transmitted RA preamble, the response is successful. The UE proceeds with 3.
- If the UE does not receive a response within the RA response window or the verification of the response fails, the response fails. In this case, if the number of RA attempts of the UE is greater than the maximum number of attempts specified by the **NRDUCellPrach.MaxPreambleTransCnt** parameter, the RA procedure fails. Otherwise, the UE makes another RA attempt.

- 3. The UE performs uplink scheduled transmission.

The UE determines the timing for scheduled transmission based on timing alignment information, and sends it to the gNodeB through Msg3.

- 4. The gNodeB assists the UE in contention resolution.

After the UE sends Msg3, a contention resolution timer of 64 ms starts. The gNodeB uses the C-RNTI on the PDCCH or the Contention Resolution Identity on the PDSCH to assist the UE in contention resolution. For details, see 3GPP TS 38.213 V16.7.0.

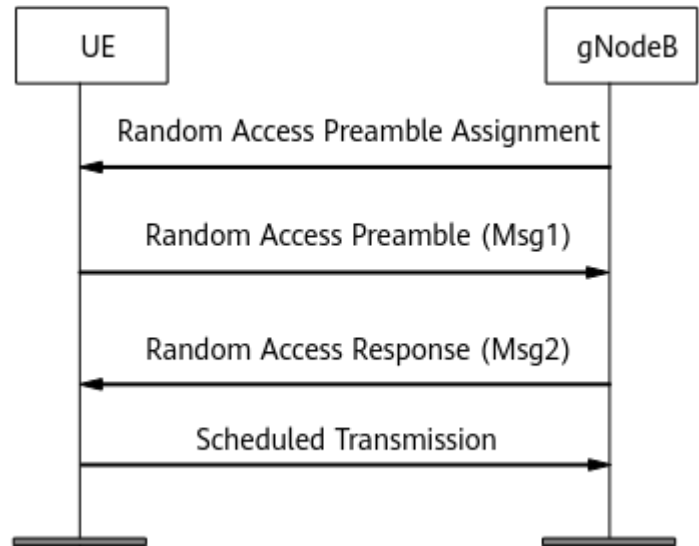
- The UE keeps monitoring the PDCCH before the timer expires. The UE considers the contention resolution successful (that is, successful access of the UE) and stops the timer if either of the following conditions is met. Otherwise, the UE does not stop the timer.
 - The UE obtains the C-RNTI over the PDCCH through Msg4. In this case, the UE stops the contention resolution timer and discards the temporary C-RNTI.
 - The UE obtains the temporary C-RNTI over the PDCCH through Msg4. The MAC packet data unit (PDU) is successfully decoded. Specifically, the *Contention Resolution Identity* IE in the MAC PDU received by the UE is the same as that carried in Msg3 sent by the UE. In this case, the UE stops the contention resolution timer and sets the temporary C-RNTI to the C-RNTI.
- If the contention resolution timer expires, the UE discards the temporary C-RNTI and considers the contention resolution failed.

Afterwards, if the number of RA attempts of the UE is greater than the maximum number of attempts specified by the **NRDUCellPrach.MaxPreambleTransCnt** parameter, the RA procedure fails. Otherwise, the UE makes another RA attempt.

4.1.3.1.2 Non-Contention-based RA Procedure

Figure 4-10 shows how non-contention-based RA works. Unlike contention-based RA, non-contention-based RA does not involve contention resolution because RA preambles are allocated by the gNodeB. Other procedures are the same for non-contention-based RA and contention-based RA. See 4.1.3.1.1 Contention-based RA Procedure.

Figure 4-10 Non-contention-based RA procedure



4.1.3.2 Procedures of RA to a gNodeB in NSA Networking

Table 4-7 describes RA triggering scenarios and mechanisms.

Table 4-7 RA triggering scenarios and mechanisms

Triggering Scenario	Scenario Description	Mechanism
RRC connection reestablishment	After detecting a radio link failure, a UE initiates RA to reestablish an RRC connection.	Contention-based RA
Handover	During a handover, a UE initiates RA in the target cell.	Non-contention-based RA is the first choice. Contention-based RA is used when dedicated preambles of the gNodeB are used up.
Downlink data arrival	When a gNodeB needs to send downlink data to an RRC_CONNECTED UE in an uplink out-of-synchronization state, the gNodeB instructs the UE to initiate RA.	Contention-based RA

Triggering Scenario	Scenario Description	Mechanism
Uplink data transmission	When an RRC_CONNECTED UE in an uplink out-of-synchronization state needs to send uplink data to a gNodeB, the UE initiates RA.	Contention-based RA
NSA access	After a UE accesses an LTE cell, the UE initiates RA in an NR cell upon the NR cell addition or change.	Non-contention-based RA is the first choice. Contention-based RA is used when dedicated preambles of the gNodeB are used up.

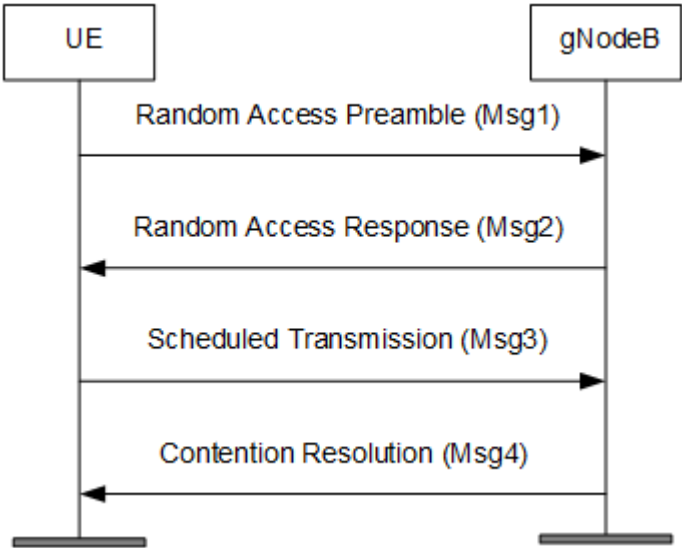
The contention-based RA procedure slightly differs from the non-contention-based RA procedure. For details about RA procedures, see section 5.1 "Random Access procedure" in 3GPP TS 38.321 V15.5.0.

4.1.3.2.1 Contention-based RA Procedure

Contention-based RA Procedure

Figure 4-11 shows the contention-based RA procedure.

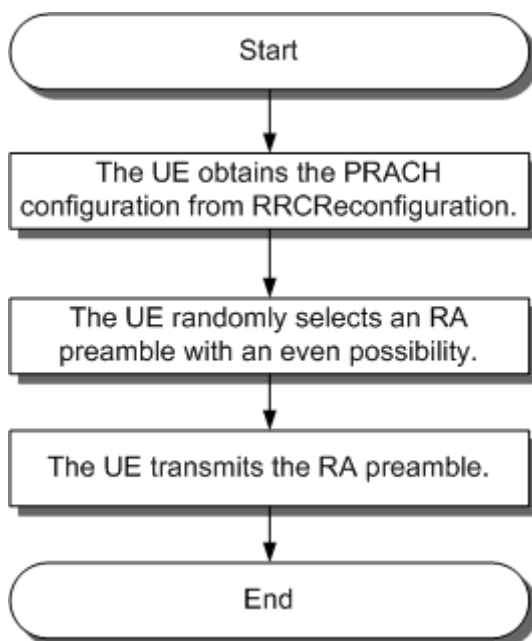
Figure 4-11 Contention-based RA procedure



The contention-based RA procedure is as follows:

1. A UE sends an RA preamble.
During RRC connection reestablishment, the UE sends an RA preamble using Msg1. Figure 4-12 shows the RA preamble transmission procedure.

Figure 4-12 RA preamble transmission procedure

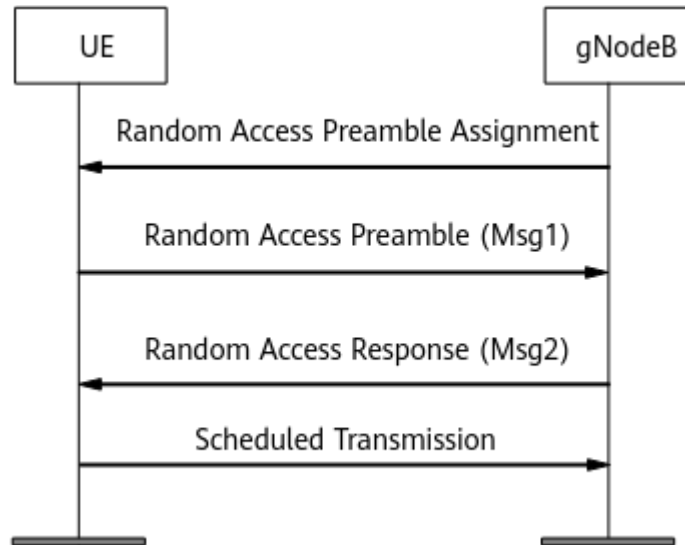


- a. The UE obtains the PRACH configuration from RRCReconfiguration.
The UE obtains the time-frequency domain position for transmitting an RA preamble based on the PRACH configuration.
- b. The UE randomly selects an RA preamble with an even possibility.
Each SSB corresponds to a unique preamble index. Therefore, before selecting an RA preamble, the UE first selects an SSB, and then determines the RA preamble based on the SSB. The UE compares the RSRP values of all SSBs in the cell with the RSRP threshold and selects an SSB whose RSRP value is greater than the threshold. The RSRP threshold is indicated by the *rsrp-ThresholdSSB* IE in 3GPP TS 38.331 V15.5.0 and specified by the **NRDUCellPrach.RsrpThldForSsbSelection** parameter. If no SSB meets the requirement, the UE selects an SSB randomly. For details, see section 5.1.2 "Random Access Resource selection" in 3GPP TS 38.321 V15.5.0.
- c. The UE sends the RA preamble.
2. The gNodeB sends an RA response.
3. The UE performs uplink scheduled transmission.
The UE determines the timing for scheduled transmission based on timing alignment information. At least the C-RNTI of the UE is transmitted.
4. The gNodeB assists the UE in contention resolution.

4.1.3.2.2 Non-Contention-based RA Procedure

Figure 4-13 shows how non-contention-based RA works. Unlike contention-based RA, non-contention-based RA does not involve contention resolution because RA preambles are allocated by the gNodeB. Other procedures are the same for non-contention-based RA and contention-based RA. See [4.1.3.2.1 Contention-based RA Procedure](#).

Figure 4-13 Non-contention-based RA procedure



4.2 Network Analysis

4.2.1 Benefits

PRACH time-frequency configurations help optimize PRACH resource utilization and improve network performance.

When the function of staggered PRACH frequency-domain resource allocation is enabled in cells with only uplink light-traffic UEs, it can reduce root sequence conflicts and increase the contention-based RA success rate ($\text{N.RA.Contention.Resolution.Succ/N.RA.Contention.Att} \times 100\%$).

In high frequency bands, when the RA preamble format is A3 (with **NRDUCellPrach.PrachConfigurationIndex** set to 74) and the cell radius (specified by **NRDUCell.CellRadius**) is set to a large value, this function improves the access performance of cell edge users (CEUs). In addition, the lower the UEs' moving speed, the higher the gains. When the cell radius is set to a small value, this function brings no gains.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

In NSA networking, when the *ServingCellConfigCommon* IE is updated, UEs re-access the network, causing transient service interruption. For details, see section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V15.5.1.

In high frequency bands, different PRACH configuration indexes (specified by the **NRDUCellPrach.PrachConfigurationIndex** parameter) correspond to different PRACH time-frequency resources. The number of time-frequency resources occupied by the PRACH has the following impacts:

- The fewer the resources occupied by the PRACH, the higher the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) and the

average uplink UE throughput (**User Uplink Average Throughput (DU)**), and the longer the access delay.

- In uplink heavy-load scenarios, the fewer the resources occupied by the PRACH, the higher the uplink PRB usage ($\text{N.PRB.Ul.Used.Avg} / \text{N.PRB.Ul.Avail.Avg}$).
- In single-UE scenarios, the fewer the resources occupied by the PRACH, the smaller the number of non-contention-based RA attempts (**N.RA.Dedicated.Att**).
- In multi-UE scenarios, the fewer the resources occupied by the PRACH, the smaller the number of non-contention-based RA attempts (**N.RA.Dedicated.Att**) and the larger the number of contention-based RA attempts (**N.RA.Contention.Att**).

After staggered PRACH frequency-domain resource allocation is enabled, network performance is affected as follows:

- When there are no UEs with heavy uplink traffic in the cell, the cell throughput may decrease.
- When there are UEs with heavy uplink traffic in the cell and there are UEs accessing a neighboring cell, interference may occur between the PUSCH and PRACH, causing the following impacts:
 - For the local cell: The uplink BLER slightly increases, and the average uplink MCS index slightly decreases. As a result, the uplink throughput slightly decreases and the PDCCH CCE usage slightly increases.
 - For neighboring cells: The number of false alarms increases. As a result, the PDCCH CCE resource overhead increases and the RA success rate decreases.

In high frequency bands, when the RA preamble format is A3 (**NRDUCellPrach.PrachConfigurationIndex** is set to **74**), a smaller number of admitted UEs and a larger number of CEUs indicate a lower or unchanged contention-based RA success rate (contention-based RA success rate = $\text{N.RA.Contention.Resolution.Succ} / \text{N.RA.Contention.Att} \times 100\%$). When the RA preamble format is A3, the following counters are affected as CEUs access the network:

- The values of the following counters may remain unchanged or decrease:
 - **RRC Setup Success Rate (CU)**
 - **NGSIG Connection Setup Success Rate (CU)**
 - **QoS Flow Setup Success Rate (CU)**
 - Non-contention-based RA success rate = $\text{N.RA.Dedicated.Msg3} / \text{N.RA.Dedicated.Att} \times 100\%$
 - Average downlink MCS index and average uplink MCS index
 - Average downlink MCS index = $(0 \times \text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times \text{N.ChMeas.PDSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PDSCH.MCS.28}) / (\text{N.ChMeas.PDSCH.MCS.0} + \text{N.ChMeas.PDSCH.MCS.1} + \text{N.ChMeas.PDSCH.MCS.2} + \dots + \text{N.ChMeas.PDSCH.MCS.28})$
 - Average uplink MCS index = $(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PUSCH.MCS.28}) / (\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \dots + \text{N.ChMeas.PUSCH.MCS.28})$

- $$\frac{N.ChMeas.PUSCH.MCS.28}{(N.ChMeas.PUSCH.MCS.0 + N.ChMeas.PUSCH.MCS.1 + N.ChMeas.PUSCH.MCS.2 + \dots + N.ChMeas.PUSCH.MCS.28)}$$
- Average downlink rank and average uplink rank
 - Average downlink rank = $(1 \times N.PDSCH.InitTbDL.Rank1 + 2 \times N.PDSCH.InitTbDL.Rank2 + 3 \times N.PDSCH.InitTbDL.Rank3 + 4 \times N.PDSCH.InitTbDL.Rank4) / (N.PDSCH.InitTbDL.Rank1 + N.PDSCH.InitTbDL.Rank2 + N.PDSCH.InitTbDL.Rank3 + N.PDSCH.InitTbDL.Rank4)$
 - Average uplink rank = $(1 \times N.PUSCH.InitTbUL.Rank1 + 2 \times N.PUSCH.InitTbUL.Rank2) / (N.PUSCH.InitTbUL.Rank1 + N.PUSCH.InitTbUL.Rank2)$
 - The values of the following counters may increase in light-load scenarios and decrease in medium- and heavy-load scenarios:
 - Downlink PRB usage
 - **N.PDCP.Vol.DL.TrfSDU.Rx** and **N.ThpVol.DL**
 - **Cell Downlink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)**
 - **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**
 - The values of the following counters may increase in light-load scenarios and fluctuate in medium- and heavy-load scenarios:
 - Uplink PRB usage
 - **N.PDCP.Vol.UL.TrfSDU.Tx** and **N.ThpVol.UL**
 - Values of the following counters may increase: **N.RA.Contention.Resp**, **N.RA.Contention.Att**, **N.RA.Dedicated.Resp**, **N.RA.Dedicated.Att**, **N.RA.Dedicated.Msg3**, **N.RA.Contention.Msg3**, and **N.RA.Contention.Resolution.Succ**.

You are advised to enable staggered PRACH frequency-domain resource allocation on the entire network or for all cells in contiguous coverage areas with root sequence conflicts. If this function is enabled only in some cells, interference may occur between the PUSCH and PRACH of the cells where this function is enabled and those where it is disabled. As a result, the uplink BLER will slightly increase for the latter, and the RA success rate will decrease for the former.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Full-bandwidth initial BWP configuration	INIT_BWP_FULL_BW_SW option of the NRDUCellIgoSwitch.BwpConfigPolicySwitch parameter	<i>5G Networking and Signaling</i>	Staggered PRACH frequency-domain resource allocation (controlled by the PRACH_FREQ_POS_STAGGER_SW option of the NRDUCellPrach.PrachAlgoExtSwitch parameter) requires full-bandwidth initial BWP configuration.
High-frequency TDD	None	None	None	None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUCCH frequency-domain position optimization	PUCCH_FREQ_POS_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Scalable Bandwidth</i>	PUCCH frequency-domain position optimization cannot work with staggered PRACH frequency-domain resource allocation (controlled by the PRACH_FREQ_POS_STAGGER_SW option of the NRDUCellPrach.PrachAlgoExtSwitch parameter).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Power saving BWP cannot work with staggered PRACH frequency-domain resource allocation (controlled by the PRACH_FREQ_POS_STAGGER_SW option of the NRDUCellPrach.PrachAlgoExtSwitch parameter).
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapabilityBasedFunctionSw parameter	<i>RedCap</i>	RedCap Introduction cannot work with staggered PRACH frequency-domain resource allocation (controlled by the PRACH_FREQ_POS_STAGGER_SW option of the NRDUCellPrach.PrachAlgoExtSwitch parameter).

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	RB number configuration for common control resources	NRDUCellCoreset.CommonCtrlResRbNum	None	- When the NRDUCellPrach.PrachConfigurationIndex parameter is set to 176, 180, or 189: - If the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB24, the value of the NRDUCell.PrachFreqStartPosition parameter can only be 65535 or range from 2 to 10. - If the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB48 or RB96, the value of the NRDUCell.PrachFreqStartPosition parameter can only be 65535 or range from 2 to 34. - When the NRDUCellPrach.PrachConfigurationIndex parameter is set to 74, 81,

RAT	Function Name	Function Switch	Reference	Description
				188, 192, or 65535: - If the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB24, the value of the NRDUCell.PrachFreqStartPosition parameter can only be 65535 or range from 2 to 16. - If the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB48 or RB96, the value of the NRDUCell.PrachFreqStartPosition parameter can only be 65535 or range from 2 to 40.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUCCH interference coordination	PUCCH_INT RF_COORD_ SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>MIMO (TDD)</i>	The start position of the PRACH frequency domain may affect the effect of PUCCH interference coordination.
High-frequency TDD	Doubled PRACH period	NRDUCellPrach.CfraPrachConfigIndex set to 188 or 74 NRDUCellPrach.PrachConfigurationIndex set to 192 or 81	<i>Intelligent Aggregate Scheduling (High-Frequency TDD)</i>	If doubled PRACH period is enabled and the maximum preamble transmission count is less than the scenario's recommended value, there is a low probability that UEs experience service drops. The recommended values are as follows: • N50 when the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter is selected • N20 when the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter is deselected

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- Basic functions of PRACH time-frequency domain configuration (preamble format and PRACH SCS configuration used by the gNodeB, configuration of the start logical ZC root sequence index, configuration of the proportion of contention-based preamble sequences, configuration of the PRACH configuration index used by the gNodeB, and configuration of the PRACH frequency-domain start position)
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Staggered PRACH frequency-domain resource allocation
All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- Basic functions of PRACH time-frequency domain configuration (preamble format and PRACH SCS configuration used by the gNodeB, configuration of the start logical ZC root sequence index, configuration of the proportion of contention-based preamble sequences, configuration of the PRACH configuration index used by the gNodeB, and configuration of the PRACH frequency-domain start position)
All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Staggered PRACH frequency-domain resource allocation
All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

In the case of high-frequency cells in a multi-carrier sector, PRACH channel resource configurations, such as the PRACH SCS and PRACH configuration index, must be the same for all the cells in the same sector. For details, see *Cell Management*. There are no such requirements in low frequency bands.

For details about the PRACH SCS supported in the current version, see [Table 4-3](#).

In a low-frequency NR TDD cell, the probability of UE access or handover failures increases because of factors such as atmospheric duct interference, if the **NRDUCellPrach.PrachConfigurationIndex** parameter is set to **65535** or **210** in one of the following scenarios: To prevent this issue and minimize the access or handover delay, it is recommended that the **NRDUCellPrach.PrachConfigurationIndex** parameter be set to **202** or **206** and resource allocation optimization for common control information (controlled by the **COMM_CTRL_INFO_ALLOC_OPT_SW** option of the **NRDUCELLPDSCH.DlPdschAlgoSwitch** parameter) be disabled in these scenarios:

- The **NRDUCell.SlotAssignment** parameter is set to **4_1_DDDSU**, and the **NRDUCell.SlotStructure** parameter is set to **SS18**.
- The **NRDUCell.SlotAssignment** parameter is set to **8_2_DDDDDDDDSUU**, the **NRDUCell.SlotStructure** parameter is set to **SS518**, and the **NRDUCell.HighSpeedFlag** parameter is set to **HIGH_SPEED**.
- The **NRDUCell.SlotAssignment** parameter is set to **8_2_DDDSUUDDDD**, the **NRDUCell.SlotStructure** parameter is set to **SS818**, and the **NRDUCell.HighSpeedFlag** parameter is set to **HIGH_SPEED**.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 4-8](#) describes the parameters used for activation.

Table 4-8 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
PRACH Configuration Index	NRDUCellPrach.PrachConfigurationIndex	Configure this parameter based on the network plan. For details, see Table 4-3 . In high-frequency TDD scenarios, it is recommended that this parameter be set to 188 or 65535 when inter-FR CA is enabled (by selecting the INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter).
CBRA Preamble Percentage	NRDUCellPrach.CbraPreamblePct	Configure this parameter based on the network plan. For details, see 4.1.1.3 Preamble Sequence Grouping .

Parameter Name	Parameter ID	Setting Notes
PRACH Frequency Start Position	NRDUCell.PrachFreqStartPosition	Configure this parameter based on the network plan. For details, see 4.1.2 Time-Frequency Resources for RA .
PRACH Algorithm Extension Switch	NRDUCellPrach.PrachAlgoExtSwitch	In low-frequency TDD, it is recommended that the logical root sequence indexes of neighboring cells be replanned to prevent PRACH root sequence conflicts. If PRACH root sequence conflicts persist after replanning, it is recommended that the PRACH_FREQ_POS_STAGGER_SW option be selected on the entire network or for all cells in contiguous coverage areas with root sequence conflicts. If this option is selected only for some cells, the impacts described in 4.2.2 Impacts may occur. Staggered PRACH frequency-domain resource allocation is only supported among cells with different PCI modulo 2 values. You are advised to properly plan PCIs for cells to be enabled with this function.

NOTE

Modifying the preceding parameters will cause the cell to restart. If the parameter values do not comply with product restrictions, the cell will fail to be activated.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Low-frequency TDD command examples

```
//Setting the PRACH configuration index (Value 65535 is used as an example, indicating that the PRACH
configuration index used by the gNodeB is automatically generated.)
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=65535;
//Setting the proportion of contention-based RA preambles
MOD NRDUCELLPRACH: NrDuCellId=0, CbraPreamblePct=255;
//Setting the start frequency-domain position of the PRACH (Value 65535 is used as an example, indicating
```

```
that the PRACH is located at the lower end of the initial BWP.)  
MOD NRDUCELL: NrDuCellId=0, PrachFreqStartPosition=65535;  
//Enabling staggered PRACH frequency-domain resource allocation when the start frequency-domain  
position of the PRACH is set to 65535  
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=PRACH_FREQ_POS_STAGGER_SW-1;
```

High-frequency TDD command examples

```
//Setting the PRACH configuration index (Value 65535 is used as an example, indicating that the PRACH  
configuration index used by the gNodeB is automatically generated.)  
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=65535;  
//Setting the proportion of contention-based RA preambles  
MOD NRDUCELLPRACH: NrDuCellId=0, CbraPreamblePct=255;  
//Setting the start frequency-domain position of the PRACH (Value 65535 is used as an example, indicating  
that the PRACH is located at the lower end of the initial BWP.)  
MOD NRDUCELL: NrDuCellId=0, PrachFreqStartPosition=65535;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

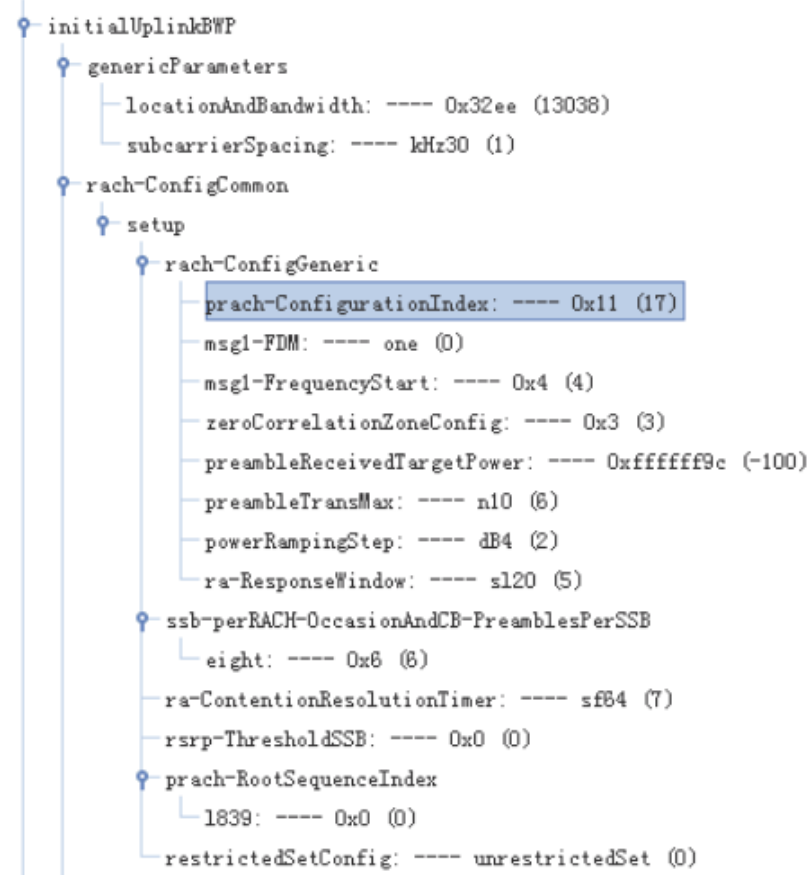
PRACH preamble configuration:

- Step 1** Start Uu signaling tracing on the MAE-Access as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
- Step 2** Enable a UE to access the network. In SA networking, check the SIB1 tracing result, as shown in [Figure 4-14](#). Check the fields in the *rach-ConfigCommon* IE in SIB1. If the value of **prach-ConfigurationIndex** is the same as the configured value, this function has taken effect.

NOTE

In NSA networking, check the LTE message RRCConnectionReconfiguration for the IEs described in SA networking.

Figure 4-14 SIB1 tracing report (SA networking)



----End

Staggered PRACH frequency-domain resource allocation: Select two cells with different PCI parity and enable UEs to access the cells. After this function is enabled, check the values of *msg1-FrequencyStart* in the *rach-ConfigGeneric* IE in SIB1 of the two cells. If the value of *msg1-FrequencyStart* for one cell is adjacent to the PUCCH at the lower end of the frequency band and the value of *msg1-FrequencyStart* for the other cell is adjacent to the PUCCH at the higher end of the frequency band, this function has taken effect.

4.4.3 Network Monitoring

Observe the following counters to monitor RA:

- Counters related to contention-based RA: **N.RA.Contention.Att**, **N.RA.Contention.Att.Max**, **N.RA.Contention.Resp**, **N.RA.Contention.Msg3**, and **N.RA.Contention.Resolution.Succ**
- Counters related to non-contention-based RA: **N.RA.Dedicated.Att**, **N.RA.Dedicated.Resp**, and **N.RA.Dedicated.Msg3**

5 RA Optimization Functions

5.1 Principles

5.1.1 PRACH Preamble Detection Threshold Increase

If the gNodeB detects a preamble from a UE, but the UE did not send a preamble, the erroneous detection of the preamble is considered as a PRACH false alarm. The false alarm probability can be reduced by increasing the preamble detection threshold through the **NRDUCellPrach.PrachThldIncreaseRate** parameter. It is recommended that this parameter be modified when the false alarm probability of a cell is higher than expected.

- A larger value of this parameter results in a lower probability of both preambles detected and false alarms. It also results in a higher probability of both miss-detection and access failures for cell-edge users (CEUs) with weak coverage.
- A smaller value of this parameter results in a higher probability of both preambles detected and false alarms. It also results in a lower probability of both miss-detection and access failures for CEUs with weak coverage.

For details about false alarms, see section 8.4.1 "PRACH False alarm probability" in 3GPP TS 38.104 V15.6.0.

PRACH false alarm detection optimization takes effect if it is enabled. For details, see [5.1.2 PRACH False Alarm Detection Optimization](#).

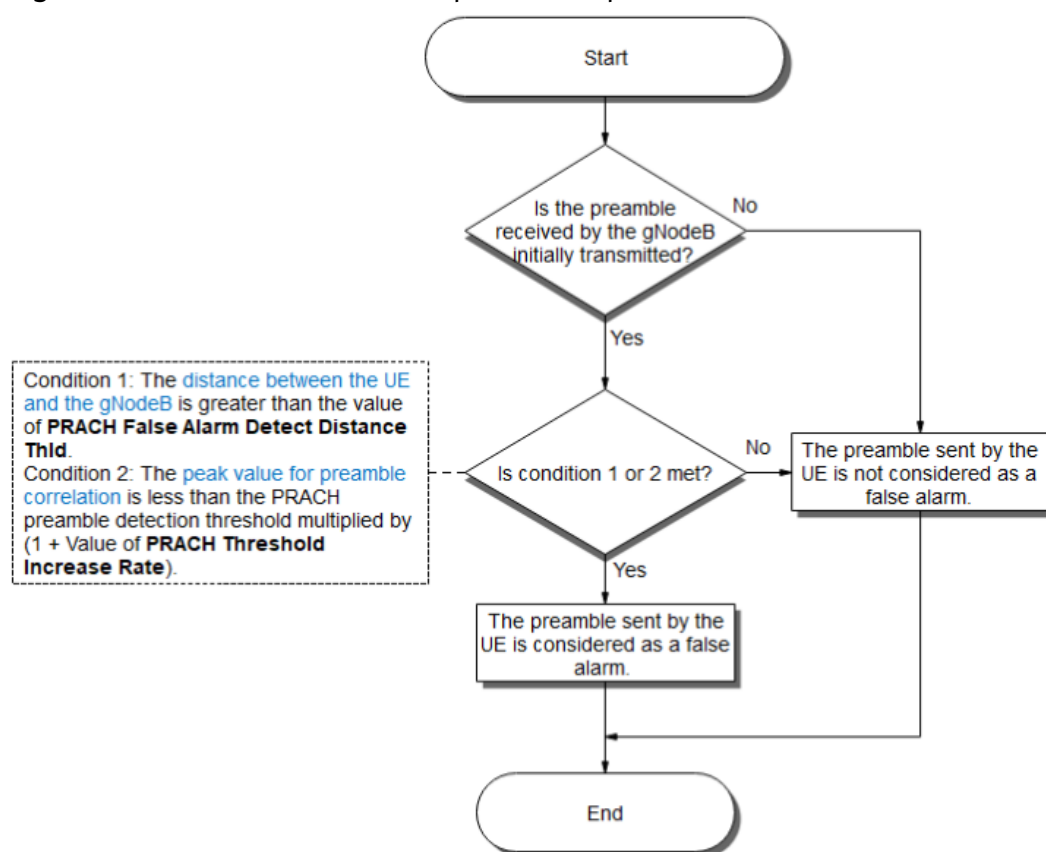
5.1.2 PRACH False Alarm Detection Optimization

If the gNodeB considers a received contention-based RA preamble as a false alarm, the gNodeB does not send the UE a Random Access Response message and does not count the preamble in related counters.

This function is controlled by the **PRACH_FALSE_ALARM_DET_ALGO_SW** option of the **NRDUCellPrach.RachAlgoOptSwitch** parameter. It is recommended that this function be enabled when the false alarm probability of a cell is higher than expected. This function takes effect only in low frequency bands.

[Figure 5-1](#) shows the PRACH false alarm detection decision mechanism.

Figure 5-1 False alarm detection optimization process



It is recommended that this function be used with the **NRDUCellPrach.PrachFalseAlmDetDisThld** and **NRDUCellPrach.PrachThldIncreaseRate** parameters specified. For details about the PRACH Threshold Increase Rate, see [5.1.1 PRACH Preamble Detection Threshold Increase](#).

5.1.3 PRACH Capability Optimization

PRACH capability optimization improves the processing capability of gNodeB baseband processing units, increasing the maximum number of UEs that are allowed to access a cell per second. This function is controlled by the **PRACH_ACCESS_CAPB_OPT_SW** option of the **NRDUCellPrach.PrachAlgoExtSwitch** parameter.

5.1.4 Narrow-Beam Probing (High-Frequency TDD)

In high-frequency TDD, narrow-beam probing is introduced. This function enables the gNodeB to select a beam range suitable for preamble reception depending on the signal quality reported from a UE, as shown in [Figure 5-2](#). This function is controlled by the **NARROW_BEAM_PROBE_SW** option of the **NRDUCellAlgoSwitch.FullChannelCovEnhSwitch** parameter and applies only to high-frequency TDD cells. With this function enabled, the beams for receiving preambles are selected as follows:

- For cell-center users (CCUs), the gNodeB receives preambles on random access channel (RACH) beams.

- For CEUs, the gNodeB receives preambles first on RACH beams throughout the coverage area. If no preambles are received, the gNodeB then uses narrow beams associated with the corresponding SSB beams. This expands the preamble reception area, since narrow beams have a longer coverage distance than RACH beams. However, this will increase access delay because narrow beams have a smaller coverage angle than RACH beams and UEs may need to increase the number of preamble transmissions to successfully access the network.

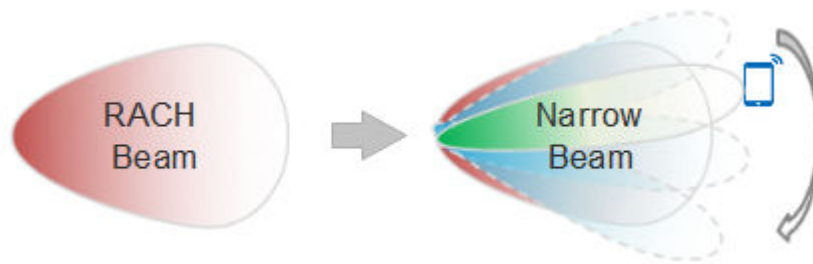
NOTE

A UE is a CCU if the SSB RSRP value in the measurement report for any of the following events is greater than or equal to the threshold specified by the **NRDUCellPrach.NarrowBeamProbeThld** parameter or a CEU if the SSB RSRP value in the measurement report for any of the following events is less than the threshold:

- Event A2 or A5 in inter-FR CA scenarios
- Event B1, A3, or A5 in non-inter-FR CA scenarios

For details about inter-FR CA, see *Carrier Aggregation*.

Figure 5-2 Narrow-beam probing



5.1.5 PRACH Coverage Enhancement (High-Frequency TDD)

In high-frequency TDD scenarios with preamble format of C2 (see [Table 4-3](#)), PRACH coverage enhancement is supported. This function can improve the PRACH detection capability and therefore increase the PRACH coverage at the cell edge. This function takes effect when the **PRACH_COV_ENH_SW** option of the **NRDUCellPrach.RachAlgoOptSwitch** parameter is selected. When the **NRDUCell.CellRadius** parameter is set to a large value, this function improves the access performance of CEUs. In addition, the lower the UEs' moving speed, the higher the gains. When the **NRDUCell.CellRadius** parameter is set to a small value, this function does not produce any gains. You are advised to enable this function in a cell where UEs move at a speed below or equal to 60 km/h.

- When the **PRACH_COV_ENH_SW** option of the **NRDUCellPrach.RachAlgoOptSwitch** parameter is selected and the **NRDUCell.CellRadius** parameter is set to a large value, the PRACH coverage area expands.
- When the **PRACH_COV_ENH_SW** option of the **NRDUCellPrach.RachAlgoOptSwitch** parameter is deselected, network performance is not affected.

5.1.6 PRACH Root Sequence Conflict Detection and Self-Optimization

PRACH preamble sequences are generated through cyclic shifts of ZC root sequences. Each root sequence can generate multiple preamble sequences. For more information, see [4.1.2 Time-Frequency Resources for RA](#).

If the frequency-domain resources of the PRACH in adjacent cells overlap and the same preamble is used, false alarms or collisions will occur during access request reception and detection in the cells. See [Figure 5-3](#) and [Figure 5-4](#).

Figure 5-3 False alarms caused by PRACH root sequences

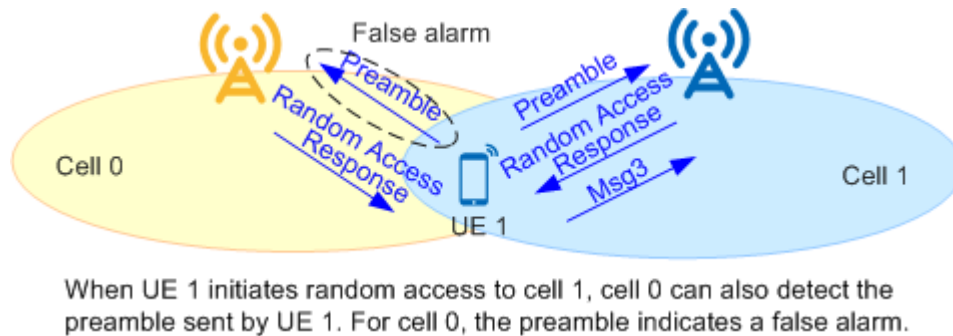
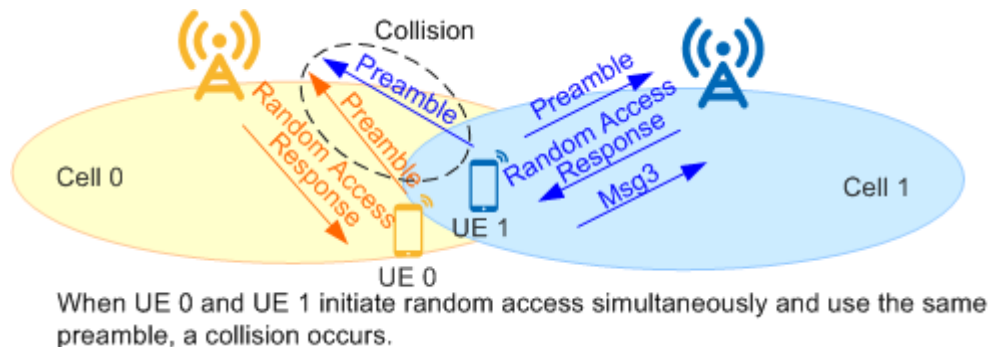


Figure 5-4 Collisions caused by PRACH root sequences



PRACH Root Sequence Conflict Detection

The MAE-Access provides the PRACH root sequence conflict detection function. This function is used to detect PRACH root sequence conflicts between the serving cell and neighboring cells managed by the MAE-Access. The MAE-Access determines that a PRACH root sequence conflict occurs when both the following conditions are met:

- The time-frequency resources of the PRACH in the serving cell overlap with those in neighboring cells.
- The serving cell and neighboring cells use the identical or partially identical root sequences.

NOTE

For details about how to enable PRACH root sequence conflict detection, see *SON Management* in *iMaster MAE Product Documentation*.

PRACH Root Sequence Conflict Self-Optimization

The MAE-Access provides the PRACH root sequence conflict self-optimization function. This function is used to resolve PRACH root sequence conflicts between the cells managed by the MAE-Access. Once a PRACH root sequence conflict between cells is detected, the MAE-Access automatically allocates a new PRACH root sequence index (specified by the **NRDUCell.LogicalRootSequenceIndex** parameter) and a new PRACH frequency-domain start position (specified by the **NRDUCell.PrachFreqStartPosition** parameter) to the conflicting cell.

NOTE

When the MAE-Access reallocates new PRACH root sequence indexes and PRACH frequency-domain start positions to cells, the cells automatically restart and become unavailable, interrupting ongoing services of UEs for a short period.

For details about how to enable PRACH root sequence conflict self-optimization, see *SON Management* in *iMaster MAE Product Documentation*.

5.1.7 Optimization of Non-Contention-based RA Beyond Cell Radius

When a UE initiates a non-contention-based RA beyond the cell radius configured on the gNodeB (by sending a dedicated preamble), the gNodeB cannot correctly identify the dedicated preamble. As a result, the non-contention-based RA procedure fails.

However, the procedure of non-contention-based RA initiated by a UE beyond the configured cell radius has been optimized to ensure successful access.

This function is controlled by the **CONTENTION_FREE_RA_OPT_SW** option of the **NRDUCellPrach.RachAlgoOptSwitch** parameter.

This function is recommended when the non-contention-based RA success rate in a cell (**N.RA.Dedicated.Msg3**) is lower than expected, and takes effect only when all of the following conditions are satisfied:

- The cell is a normal low-speed cell (this function is not supported in high-speed railway/hyper cell/Cell Combination/CA scenarios).
- The distance between the UE and the base station is greater than the cell radius but less than twice the cell radius.

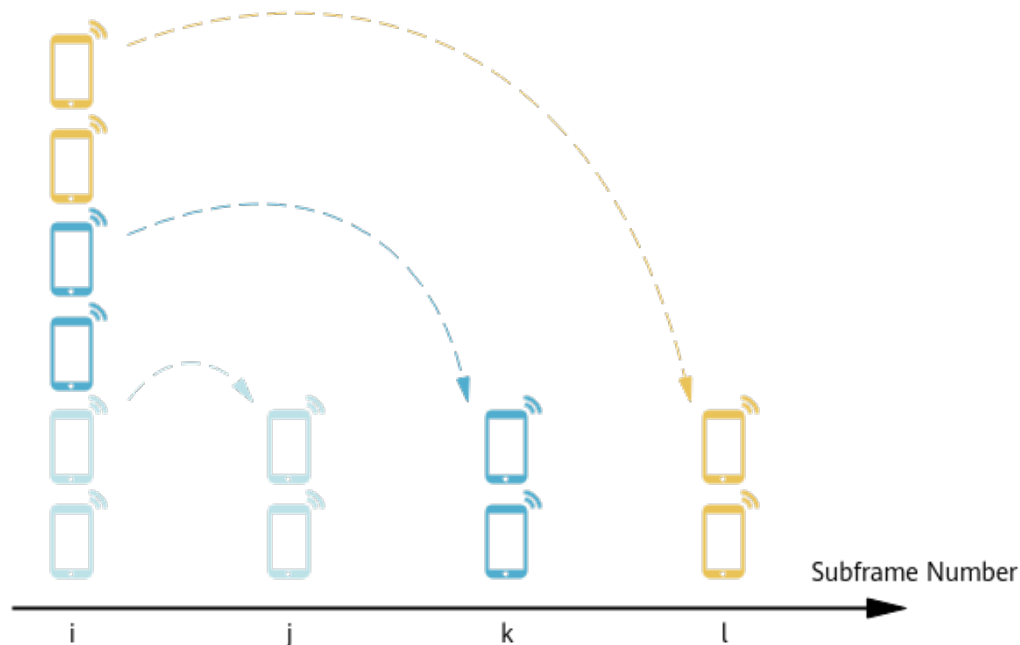
5.1.8 Backoff Control

During contention-based RA, preamble collisions may occur when multiple UEs send preambles in the same PRACH time-frequency domain position, causing RA failures. The UEs then need to retransmit the preambles. To reduce the probability of preamble collisions during preamble retransmission, backoff control can be used during RA.

After backoff control is enabled, the gNodeB informs the UE of a backoff index through the RA response message in the event of preamble collisions. Each backoff index corresponds to a maximum backoff time. When retransmitting a preamble, a UE randomly selects a value from 0 to the maximum backoff time as the backoff time. The UE retransmits the preamble after the backoff time expires. [Figure 5-5](#) shows backoff control. When multiple UEs collide in subframe *i*,

backoff control enables each UE to randomly select a backoff time and retransmit preambles in subframes j, k, and l, separately. For details about backoff control, see section 5.1 "Random Access procedure" in 3GPP TS 38.321 V17.0.0.

Figure 5-5 Backoff control



The **BACKOFF_SW** option of the **NRDUCellPrach.RachAlgoOptSwitch** parameter specifies whether to enable backoff control. This function takes effect only in low-frequency scenarios. It is recommended that this function be enabled when contention-based RA is frequent.

UEs are not subject to backoff control in any of the following scenarios:

- The UE transmits a preamble during the first contention-based RA procedure.
- The UE transmits a preamble in a non-contention-based RA procedure.

Whether this function takes effect is subject to the SSB and PRACH association period.

- When the association period is less than or equal to 40 ms, the gNodeB controls the backoff index based on the association period.
- This function does not take effect when the association period is longer than 40 ms.

NOTE

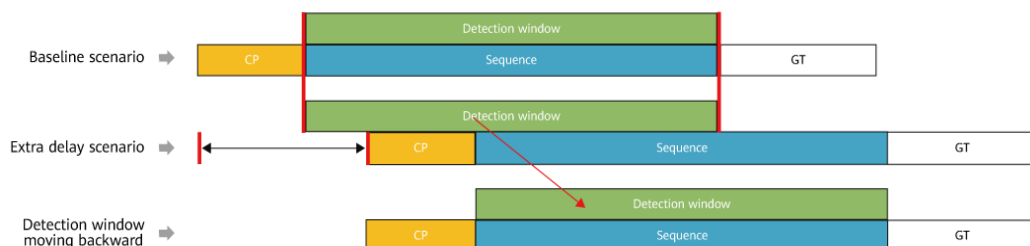
There is a mapping relationship between SSB indexes and PRACH resources, and different SSB indexes are mapped to PRACH resources of different time-frequency domain positions. The SSB and PRACH association period is defined as the time when PRACH resources corresponding to all SSB indexes are traversed, that is, SSB and PRACH association period = PRACH time-frequency resource configuration period x (Number of SSBs/Number of SSB indexes associated with each PRACH time-frequency resource). The number of SSBs should be an element in the set {1, 2, 4, 8}. If the actual number of SSBs in the system does not belong to the set, the minimum element in the set that is greater than the actual number of SSBs in the system is used. For example, assuming that the number of SSBs is 7, the PRACH time-frequency resource configuration period is 5 ms, and each PRACH time-frequency resource may be associated with 4 SSB indexes, when the association period is calculated, the number of SSBs should be 8. In this scenario, the association period between the SSB and the PRACH is 10 ms.

For details about the mapping between SSB indexes and PRACH resources, see section 8.1 "Random access preamble" in 3GPP TS 38.213 V17.2.0.

5.1.9 PRACH Fixed Delay Compensation

During RA, if the air interface delay exceeds the delay allowed by the PRACH format, the preamble sequence that falls into the PRACH detection window is incomplete. See [Figure 5-6](#). Consequently, the detection performance deteriorates. As a result, RA fails. To eliminate the impact of additional air interface delay introduced by external devices on the PRACH detection performance, the PRACH fixed delay compensation function is introduced. This function is controlled by the **NRDUCellTrpAlgo.FixedDelayCompensation** parameter.

Figure 5-6 Impact of air interface delay on PRACH detection performance



Based on the known fixed air interface delay, the PRACH detection window is moved backward to offset the impact of the air interface delay. This ensures that the preamble sequence falling into the detection window is complete, thereby avoiding the impact of the air interface delay on the detection performance and ensuring normal UE access. This function is recommended to maintain access performance in the face of additional air interface delay from external devices connected to RRUs and pRRUs.

NOTE

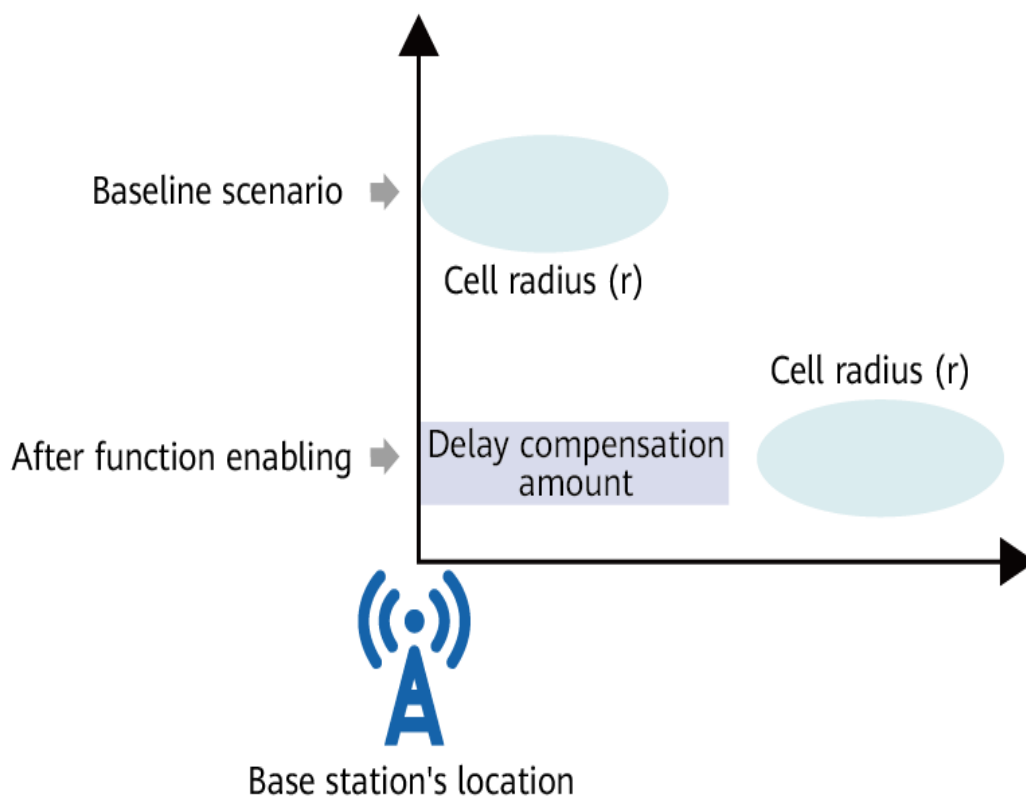
- The fixed delay compensation function requires that the delay to be compensated (that is, the extra air interface delay caused by other external devices) be a fixed value which is measurable. If the extra air interface delay introduced by the external device changes, the fixed delay compensation value needs to be reconfigured accordingly.
- The delay compensation amount in this function denotes the total uplink and downlink delay.

The fixed delay compensation function shifts the air interface coverage distance of PRACH format C2 from 0 to 9.65 km to (X to $X+9.65$) km (maximum value of X).

44.2 km). **Figure 5-7** shows the changes in the cell radius. After this function is enabled, the cell coverage area does not include the coverage distance caused by air interface delay. The cell coverage area can be set to a smaller value to facilitate root sequence planning.

In addition to the cell radius, the air interface distance equivalent to the fixed delay needs to be added to the actual coverage distance. Therefore, the timeslot structure needs to be adjusted based on the actual air interface coverage distance. For example, if the cell radius is 6 km (assuming that the original timeslot structure is SS4) and the air interface coverage distance corresponding to the delay compensation configuration is 14 km, the actual air interface coverage distance is 20 km. Based on the timeslot structure requirements, the air interface coverage distance must be set to **SS6**. For details about the timeslot structure, see *Standards Compliance*.

Figure 5-7 Impact of fixed delay compensation on the cell coverage area



Currently, the granularity of fixed delay compensation parameters is 1 μ s. If the extra air interface delay measurement introduced on the live network is more refined, it is recommended that the value be rounded down. For example, if the delay to be compensated on the live network is 9.6 μ s, the **NRDUCellTrpAlgo.FixedDelayCompensation** parameter needs to be set to **Delay_5_9**.

To ensure that CEUs can access the network, a margin needs to be reserved for the cell radius in addition to the corresponding delay compensation value. The margin can be configured based on **Table 5-1**. (For example, in **DELAY_5_9**, if the required cell radius is 500 m, the recommended cell radius is 1250 m.)

If the delay to be compensated exceeds the maximum fixed delay compensation value 291 μ s, the fixed delay compensation value needs to be at the level of **DELAY_291_357**. In addition, the cell radius also needs to be adjusted based on the total system delay (that is, the sum of the extra air interface delay and the existing air interface delay) to ensure UE access. If the required total system delay changes, the cell radius needs to be adjusted accordingly. **Table 5-1** lists the adjustment for the cell radius (**NRDUCell.CellRadius**). PRACH fixed delay compensation is supported in low-frequency TDD with a preamble format of 0 or C2 (see **Table 4-3**). For preamble format C2, only slot assignment 4:1 is supported. When the preamble format is **0**, the **NRDUCellTrpAlgo.FixedDelayCompensation** parameter can only be set to a value between **DELAY_5_9** and **DELAY_90_94**.

Table 5-1 Recommended cell radius based on the total system delay

Configured Delay Compensation Level	Total Delay to Be Compensated (Unit: μ s)	Cell Radius Configuration (Unit: m)	Reserved Cell Radius Margin (Unit: m)	Preamble Format
DELAY_5_9 ~ DELAY_90_94	5 to 94.9	Configured based on UE distribution	750	C2, 0
DELAY_95_108 ~ DELAY_277_290	94.9 to 290	Configured based on UE distribution	2100	C2
DELAY_291_357	291 to 299	1300	N/A	C2
	299 to 301	1500		
	301 to 304	2000		
	304 to 310	2900		
	310 to 321	4600		
	321 to 357	9650		

5.1.10 NSA CBRA Statistics Optimization

In the current version, RA optimization refers to optimized counter measurement for contention-based RA. In NSA cells, optimized counter measurement for contention-based RA is enabled by selecting the **NSA_CBRA_STATISTICS_OPT_SW** option of the **gNBOamParam.StatisticsStrategySwitch** parameter. It is recommended that this function be enabled in light-load scenarios where the probability of root sequence conflicts is high and the number of RA times is small.

After this function takes effect, when RA is triggered on the gNodeB side in NSA networking, a timer with the same length as the T304 timer is started for measurement of contention-based RA performance counters. The counter values after timer expiration are not measured. For details about timer triggering

scenarios, see descriptions of counters in [Table 5-2](#) in *gNodeBFunction Performance Counter Summary*.

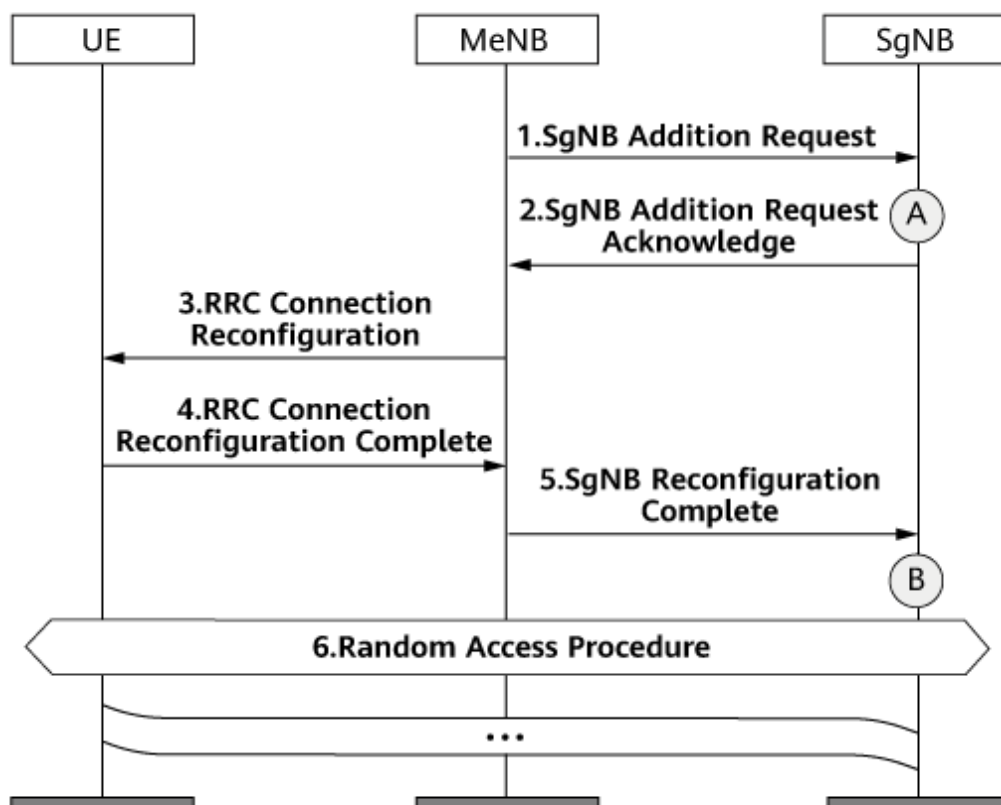
[Table 5-2](#) describes the counters related to contention-based RA. PRACH false alarms in the serving cell affect the measurement of the counters described in [Table 5-2](#), thereby affecting the contention-based RA success rate (**N.RA.Contention.Resolution.Succ/N.RA.Contention.Att**).

Table 5-2 Counters related to contention-based RA

Counter ID	Counter Name	Counter Description
1911816542	N.RA.Contention.Att	Number of random preamble receptions
1911820708	N.RA.Contention.Resp	Number of random preamble responses
1911820710	N.RA.Contention.Msg3	Number of Msg3 receptions during contention-based RA
1911816544	N.RA.Contention.Resolution.Succ	Number of successful contention resolutions

[Figure 5-8](#) shows an SgNB addition procedure where a timer is triggered. The gNodeB triggers a timer with the same length as the T304 timer at point B. In this case, the measurement of counters related to contention-based RA starts at point A and ends when the timer triggered at point B expires.

Figure 5-8 SgNB addition procedure



5.1.11 UE E-UTRAN Capability Query Optimization

During RA of a UE, the base station performs full query of E-UTRAN capabilities of the UE. The size of the capability message reported by the UE is usually too large. Especially in weak coverage scenarios, the overly large size of the message causes the RA to fail, due to a relatively high BLER.

UE E-UTRAN capability query optimization has been introduced to increase the RA success rate. The *requestedFrequencyBands-r11* IE can now be included in UE capability query messages to specify the E-UTRAN CA band combinations to be queried. Using this IE, the base station no longer needs to perform full query of E-UTRAN capabilities, and therefore the UE capability message size decreases. This function resolves RA failures caused by an excessively high BLER of UE capability messages in weak coverage scenarios. This function is controlled by the **UE_EUTRAN_CAPB_ENQUIRY_OPT_SW** option of the **gNodeBParam.ProcessSwitch** parameter.

5.1.12 UE Random Access Control Based on Cell Radius

This function forbids the access of certain UEs, such as those beyond a country's border.

This function is controlled by the **ACCESS_BYND_CELL_RAD_PROHIB_SW** option of the **NRDUCellPrach.PrachAlgoExtSwitch** parameter.

This function is implemented as follows:

1. After receiving a random access preamble from a UE, the gNodeB calculates the distance between the UE and the gNodeB based on the timing advance (TA) of the preamble.

2. If the distance is greater than the cell radius specified by the **NRDUCell.CellRadius** parameter, the gNodeB sends an RRC connection rejection message to the UE upon receiving an RRC connection request, prohibiting the UE from accessing the cell.

The function of "optimized measurement of RRC connection setup success rate based on cell radius" enables the RRC connection setup success rate to be measured more accurately. This function is controlled by the **RRC_SETUP_SUCC_RADIUS_OPT_SW** option of the **NRCCellFeatureSwitch.CounterConfigSwitch** parameter. After this function is enabled, the measurement of RRC connection setup success rate excludes UEs whose cell radius exceeds the limit.

5.2 Network Analysis

5.2.1 Benefits

Table 5-3 describes the gains provided by the sub-functions of RA optimization.

Table 5-3 Gains of the sub-functions of RA optimization

Sub-function Name	Benefits
PRACH preamble detection threshold increase	This function is controlled by the NRDUCellPrach.PrachThldIncreaseRate parameter. It is recommended that this parameter be adjusted when the false alarm probability of a cell is higher than expected. - A larger value of this parameter results in a lower probability of both preambles detected and false alarms. It also results in a higher probability of both miss-detection and access failures for cell-edge users (CEUs) with weak coverage. - A smaller value of this parameter results in a higher probability of both preambles detected and false alarms. It also results in a lower probability of both miss-detection and access failures for CEUs with weak coverage.
PRACH false alarm detection optimization	After this function is enabled, the probability that contention-based RA preambles are detected as false alarms decreases and the RA success rate increases.
PRACH capability optimization	After this function is enabled, the maximum number of UEs that are allowed to access a cell per second increases.

Sub-function Name	Benefits
Narrow-beam probing	After this function is enabled, the coverage distance of narrow beams is better than that of RACH beams. Therefore, the preamble reception range is expanded and the access success rate increases.
PRACH coverage enhancement	When this function is enabled and the NRDUCell.CellRadius parameter is set to a large value, the access performance of CEUs improves. In addition, the lower the UEs' moving speed, the higher the gains. When the NRDUCell.CellRadius parameter is set to a small value, there are no gains. You are advised to enable this function in a cell where UEs move at a speed below or equal to 60 km/h.
Optimization of non-contention-based RA beyond cell radius	After this function is enabled, when a UE initiates a non-contention-based RA beyond the cell radius configured on the gNodeB, the RA success rate increases if the following conditions are met: • The cell is a normal low-speed cell (this function is not supported in high-speed railway/hyper cell/Cell Combination/CA scenarios). • The distance between the UE and the base station is greater than the cell radius but less than twice the cell radius. If optimization of non-contention-based RA beyond cell radius is enabled when conditions are not met, this function does not take effect and does not have positive or negative impacts.
Backoff control	After this function is enabled: • When the SSB and PRACH association period is less than or equal to 40 ms, UEs that fail contention resolution initiate RA again after a period of time. This relieves RACH congestion. • When the SSB and PRACH association period is greater than 40 ms, backoff control does not take effect, and does not have positive or negative gains.
PRACH fixed delay compensation	This function is controlled by the NRDUCellTrpAlgo.FixedDelayCompensation parameter. It is recommended that this parameter be set based on the actual air interface delay which includes the extra air interface delay caused by RRUs being connected to external devices. When the actual delay is consistent with the delay compensation value specified by this parameter: • The preamble detection performance can be guaranteed and target UEs at the cell edge can access the network. • The impact of air interface delay on the coverage distance does not need to be considered. The cell radius (NRDUCell.CellRadius) can still be set to a small value, which facilitates root sequence planning.

Sub-function Name	Benefits
NSA CBRA statistics optimization	When the number of RA times is large, the measured values of counters related to contention-based RA slightly decrease or remain unchanged, and the RA success rate slightly increases or remains unchanged.
UE E-UTRAN capability query optimization	After this function is enabled, the size of UE capability messages reduces and the number of UE capability segments decreases accordingly. In weak coverage scenarios, the BLER of UE capability messages decreases, and the probability of QoS flow setup failures also decreases.
UE random access control based on cell radius	After this function is enabled, contention-based random access from UEs whose distance to the base station exceeds the cell radius is forbidden.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

[Table 5-4](#) describes the gains provided by the sub-functions of RA optimization.

Table 5-4 Network impacts of RA optimization

Sub-function Name	Network Impact
PRACH preamble detection threshold increase	For details about the impact of the NRDUCellPrach.PrachThldIncreaseRate parameter on the network after this function is enabled, see 5.2.1 Benefits .
PRACH false alarm detection optimization	After this function is enabled, contention-based RA preambles may be incorrectly considered as false alarms. This may increase the RA delay and service drop rate, and decrease the handover success rate (only in the case of handovers during which the RA procedure switches from non-contention-based to contention-based). In addition, false alarm detection slightly increases the CPU usage.

Sub-function Name	Network Impact
PRACH capability optimization	After this function is enabled, the contention-based RA success rate decreases in scenarios where contention-based preamble resources are sufficient but dedicated preamble resources are insufficient. - The values of the following counters may decrease when contention-based resources are sufficient but non-contention-based resources are insufficient: N.RA.Contention.Att, N.RA.Contention.Resp, N.RA.Contention.Msg3, and N.RA.Contention.Resolution.Succ. - The contention-based RA success rate (N.RA.Contention.Resolution.Succ/N.RA.Contention.Att) may decrease. - The values of the following counters may increase when contention-based access resources are sufficient but non-contention-based access resources are insufficient: N.RA.Dedicated.Att, N.RA.Dedicated.Resp, and N.RA.Dedicated.Msg3
Narrow-beam probing	After this function is enabled, the access coverage radius increases for CEUs under weak coverage. However, this will increase access delay because narrow beams have a smaller coverage angle than RACH beams and UEs may need to increase the number of preamble transmissions to successfully access the network. After this function is enabled, if the maximum number of preamble transmissions (specified by the NRDUCellPrach.MaxPreambleTransCnt parameter) is too small, the narrow beams may not cover CEUs, causing access failures. After this function is enabled, if a CEU accesses the network: - The values of the following counters may increase in light-load scenarios and decrease in medium- and heavy-load scenarios: - Downlink PRB usage N.PDCP.Vol.DL.TrfSDU.Rx and N.ThpVol.DL Cell Downlink Average Throughput (DU) and Cell Uplink Average Throughput (DU) User Uplink Average Throughput (DU) and User Downlink Average Throughput (DU) - The values of the following counters may increase in light-load scenarios and fluctuate in medium- and heavy-load scenarios: - Uplink PRB usage N.PDCP.Vol.UL.TrfSDU.Tx and N.ThpVol.UL

Sub-function Name	Network Impact
PRACH coverage enhancement	After this function is enabled, if a CEU accesses the network: • The values of the following counters may decrease or remain unchanged: – RRC Setup Success Rate (CU) – NGSIG Connection Setup Success Rate (CU) – QoS Flow Setup Success Rate (CU) • The values of the following counters may increase in light-load scenarios and decrease in medium- and heavy-load scenarios: – Downlink PRB usage – N.PDCP.Vol.DL.TrfSDU.Rx and N.ThpVol.DL – Cell Downlink Average Throughput (DU) and Cell Uplink Average Throughput (DU) – User Uplink Average Throughput (DU) and User Downlink Average Throughput (DU) • The values of the following counters may increase in light-load scenarios and fluctuate in medium- and heavy-load scenarios: – Uplink Resource Block Utilizing Rate – N.PDCP.Vol.UL.TrfSDU.Tx and N.ThpVol.UL
PRACH root sequence conflict detection and self-optimization	After this function is enabled, if the PRACH frequency-domain start position of the local cell is replanned, the cell throughput may decrease. In addition, when there are UEs accessing a neighboring cell, interference may occur between the PUSCH and PRACH, causing the following impacts: • For the local cell: The uplink IBLER slightly increases, and the average uplink MCS index slightly decreases. As a result, the uplink throughput slightly decreases and the PDCCH CCE usage slightly increases. • For neighboring cells: The number of false alarms increases. As a result, the PDCCH CCE resource overhead increases and the RA success rate decreases.

Sub-function Name	Network Impact
Optimization of non-contention-based RA beyond cell radius	Enabling this function may have the following impacts on KPIs: • The number of RRC_CONNECTED UEs increases. • The access or handover delay of UEs beyond the cell radius is longer than that of common UEs. As a result, the overall access or handover delay may increase. • If coverage for UEs located beyond the cell radius is poor, the following apply: – If the UEs successfully access the network, uplink and downlink user experience deteriorates and the service drop rate may increase. – If the UE access or handover fails, the access or handover success rate decreases.
Backoff control	After this function is enabled, if the SSB and PRACH association period is less than or equal to 40 ms, the UE access delay and handover delay may increase. If the SSB and PRACH association period is greater than 40 ms, this function has no impact on the network.
PRACH fixed delay compensation	After this function is enabled, it has the following impacts depending on the consistency between the actual delay and the specified delay compensation value: • When the actual delay is consistent with the specified delay compensation value, UEs that fail to access the network in the target coverage area due to extra air interface delay can successfully access the network. • If the actual delay is inconsistent with the specified delay compensation value, some UEs may fail to access the target cell. As a result, accessibility-related indicators of the cell deteriorate.
NSA CBRA statistics optimization	It is recommended that this function be enabled when the root sequence conflict probability is high and the RA frequency is low. When there are SUL cells or Hyper cells, the rise in RA success rate may decrease after this function is enabled.
UE E-UTRAN capability query optimization	After this function is enabled, UEs that do not support the <i>requestedFrequencyBands-r11</i> IE (which has been added to UE capability query messages) will experience issues such as unexpected service drops, CA failures, or performance deterioration.

Sub-function Name	Network Impact
UE random access control based on cell radius	This function enables the base station to detect UE access beyond the cell radius, which increases the consumption of physical resources. Since contention-based random access is prohibited for UEs whose distance to the base station exceeds the cell radius, the random access success rate and RRC connection setup success rate may decrease, and the cell traffic, cell rate, and PRB usage may decrease. In addition, enabling "optimized measurement of RRC connection setup success rate based on cell radius" can prevent the RRC connection setup success rate from decreasing.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE random access control based on cell radius	ACCESS_BYND_CELL_RAD_PROHIB_SW option of the NRDUCellPrach.PrachAlgoExtSwitch parameter	<i>Random Access Control</i>	"Optimized measurement of RRC connection setup success rate based on cell radius" takes effect only after "UE random access control based on cell radius" is enabled.

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed Railway Superior Experience cannot work with PRACH false alarm detection optimization (controlled by the PRACH_FALSE_ALARM_DET_ALGO_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter).
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed Railway Superior Experience cannot work with optimization of non-contention-based RA beyond cell radius (controlled by the CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter).
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Optimization of non-contention-based RA beyond cell radius (controlled by the CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter) cannot work with Hyper Cell (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Optimization of non-contention-based RA beyond cell radius (controlled by the CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter) cannot work with Cell Combination (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE).
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of NRDUCellRachAlgoSwitch.CaAlgoSwitch	<i>Carrier Aggregation</i>	Downlink intra-band CA cannot work with optimization of non-contention-based RA beyond cell radius (controlled by the CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter).
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellRachAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Downlink intra-FR inter-band CA cannot work with optimization of non-contention-based RA beyond cell radius (controlled by the CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter).

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	When narrow-beam probing (controlled by the NARROW_BEAM_PROBE_SW option of the NRDUCellAlgoSwitch.FullChannelCoverageEnhSwitch parameter) is enabled, the NRDUCellTrpMmwavBeam.CoverageScenario parameter cannot be set to SCENARIO_103 .
Low-frequency TDD High-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-FR CA is mutually exclusive with optimization of non-contention-based RA beyond cell radius (controlled by the CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter) and cannot be enabled together.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	FWA mmWave coverage extension	NRDUCell.CellRadius	<i>FWA</i>	PRACH coverage enhancement (controlled by the PRACH_COV_ENH_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter) and FWA mmWave coverage extension (with NRDUCell.CellRadius set to a value greater than 2410 and less than or equal to 5000) are mutually exclusive and cannot be enabled together.
Low-frequency TDD	Backoff control	BACKOFF_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter	<i>Random Access Control</i>	Backoff control is mutually exclusive with PRACH false alarm detection optimization (controlled by the PRACH_FALSE_ALARM_DET_ALGO_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter) and cannot be enabled together.

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	For multiple TRPs of a distributed massive MIMO cell, the fixed delay compensation function takes effect only based on the fixed delay configured for the master TRP.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	For a hyper cell (specified by HYPER_CELL), if the delay from the UE to the master and neighboring TRPs is large (for example, in overshoot coverage scenarios), that is, the difference between the fixed delay compensation configurations of the master and neighboring TRPs is large, UE performance is affected.

5.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

UE random access control based on cell radius is supported only by macro base stations.

Boards

- PRACH preamble detection threshold increase
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- PRACH false alarm detection optimization
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

- PRACH capability optimization
All NR-capable main control boards and the UBBPg series BBPs support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Narrow-beam probing
All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- PRACH coverage enhancement
All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- PRACH root sequence conflict detection and self-optimization
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Optimization of non-contention-based RA beyond cell radius
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Backoff control
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- PRACH fixed delay compensation
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- NSA CBRA statistics optimization
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- UE E-UTRAN capability query optimization
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- UE random access control based on cell radius
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- PRACH preamble detection threshold increase
All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

- PRACH false alarm detection optimization
All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- PRACH capability optimization
All NR TDD-capable 1T1R, 2T2R, 32T32R, and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Narrow-beam probing
All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.
- PRACH coverage enhancement
All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.
- PRACH root sequence conflict detection and self-optimization
All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Optimization of non-contention-based RA beyond cell radius
All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Backoff control
All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- PRACH fixed delay compensation
All NR-capable RRUs working on low frequency bands support this function. For details, see the technical specifications of RRUs in *3900 & 5900 Series Base Station Product Documentation*.
- NSA CBRA statistics optimization
All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- UE E-UTRAN capability query optimization
All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- UE random access control based on cell radius
All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Cells

The **NRDUCellTrp.TxRxMode** parameter must be set to **1T1R**, **2T2R**, **4T4R**, or **8T8R** for an NR cell enabled with fixed delay compensation.

When the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU** for an NR cell, PRACH coverage enhancement (high-frequency TDD) does not take effect.

When the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU** for an NR cell, narrow-beam probing (controlled by the **NARROW_BEAM_PROBE_SW** option of the **NRDUCellAlgoSwitch.FullChannelCovEnhSwitch** parameter) cannot be enabled.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-5](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-5 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
PRACH Algorithm Extension Switch	NRDUCellPrach.PrachAlgoExtSwitch	It is recommended that the PRACH_ACCESS_CAPB_OPT_SW option be selected for low-frequency TDD 1T1R and 2T2R cells served by UBBPg series boards.
PRACH Threshold Increase Rate	NRDUCellPrach.PrachThldIncreaseRate	Configure this parameter based on the network plan. For details, see 5.1.1 PRACH Preamble Detection Threshold Increase .

Parameter Name	Parameter ID	Setting Notes
Random Access Algorithm Opt Switch	NRDUCellPrach.RachAlgoOptSwitch	- Select the CONTENTION_FREE_RA_OPT_SW option when many UEs are located beyond the cell radius. - Select the PRACH_FALSE_ALARM_DET_ALGO_SW option when the false alarm probability of a cell is higher than expected. - In high-frequency TDD, it is recommended that the PRACH_COV_ENH_SW option of this parameter be selected to enable PRACH coverage enhancement for cells where UEs move at a speed less than or equal to 60 km/h. - If backoff control is required when contention-based RA is frequent, select the BACKOFF_SW option.
PRACH False Alarm Detect Distance Thld	NRDUCellPrach.PrachFalseAlmDetDisThld	You are advised to set this parameter when the false alarm probability of a cell is higher than expected and PRACH false alarm detection optimization is enabled. Configure this parameter based on the network plan. For details, see 5.1.2 PRACH False Alarm Detection Optimization.
Full Channel Coverage Enhancement Switch	NRDUCellAlgoSwitch.FullChannelCovEnhSwitch	Select the NARROW_BEAM_PROBE_SW option when narrow-beam probing is required in high-frequency TDD.
Narrow Beam Probe Threshold	NRDUCellPrach.NarrowBeamProbeThld	In high-frequency TDD scenarios: - If NRDUCellPrach.PrachConfigurationIndex is set to 176, 180, or 189, you are advised to use the default value. - If the NRDUCellPrach.PrachConfigurationIndex parameter is set to 188 or 65535, you are advised to set this parameter to -112.

Parameter Name	Parameter ID	Setting Notes
Maximum Preamble Transmission Count	NRDUCellPrach. MaxPreambleTransCnt	In high-frequency TDD scenarios: - If the NARROW_BEAM_PROBE_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter is selected and the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwav-Beam.BeamPerformanceSw parameter is deselected, you are advised to use the default value. - If the NARROW_BEAM_PROBE_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter is selected and the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwav-Beam.BeamPerformanceSw parameter is also selected, you are advised to set this parameter to N20.
Fixed delay compensation	NRDUCellTrpAlgo. FixedDelayCompensation	By default, this function is disabled (the default value is DELAY_0_4). When external devices introduce extra air interface delay, set this parameter based on the introduced fixed delay.
Statistics Strategy Switch	gNBOamParam.S tatisticsStrategy-Switch	In NSA networking, it is recommended that the NSA_CBRA_STATISTICS_OPT_SW option be selected to enable NSA CBRA statistics optimization in light-load scenarios where the RA frequency is low and the probability of root sequence conflicts is high.
Process Switch	gNodeBParam.Pr ocessSwitch	It is recommended that the UE_EUTRAN_CAPB_ENQUIRY_OPT_SW option be selected to enable UE E-UTRAN capability query optimization in weak coverage scenarios.
PRACH Algorithm Extension Switch	NRDUCellPrach.P rachAlgoExtSwitch	If accurate control of cell coverage is required, select the ACCESS_BYND_CELL_RAD_PROHIB_SW option to enable UE random access control based on cell radius. This function is not recommended in other scenarios.

Parameter Name	Parameter ID	Setting Notes
Counter Configuration Switch	NRCCellFeatureSwitch.CounterConfigSwitch	If the RRC connection setup success rate needs to be optimized, specifically, by excluding UEs whose cell radius exceeds the limit, select RRC_SETUP_SUCC_RADIUS_OPT_SW to enable "optimized measurement of RRC connection setup success rate based on cell radius".

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Low-frequency TDD command examples

```
//Setting the percentage by which the PRACH preamble detection threshold increases (0% as an example)
MOD NRDUCELLPRACH: NrDuCellId=0, PrachThldIncreaseRate=0;
//Enabling PRACH false alarm detection optimization (The PRACH preamble detection threshold and
PRACH false alarm detection distance threshold must be configured first.)
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch= PRACH_FALSE_ALARM_DET_ALGO_SW-1;
//Setting the distance threshold for PRACH false alarm detection when PRACH false alarm detection
optimization is enabled (1000 m as an example)
MOD NRDUCELLPRACH: NrDuCellId=0, PrachFalseAlmDetDisThld=1000;
//Enabling PRACH capability optimization
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=PRACH_ACCESS_CAPB_OPT_SW-1;
//Enabling non-contention-based RA optimization
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch=CONTENTION_FREE_RA_OPT_SW-1;
//Enabling backoff control
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch=BACKOFF_SW-1;
//Enabling the PRACH fixed delay compensation function (assuming that the slot configuration is 4:1, the
compensation level is DELAY_55_59, and the special timeslot is SS6)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, FrequencyBand=N41,
SlotAssignment=4_1_DDDSU, SlotStructure=SS6;
MOD NRDUCELLTRPALGO: NrDuCellTrpId=0, FixedDelayCompensation=DELAY_55_59;
//Enabling NSA CBRA statistics optimization
MOD GNBOAMPARAM: StatisticsStrategySwitch=NSA_CBRA_STATISTICS_OPT_SW-1;
//Enabling UE E-UTRAN capability query optimization
MOD GNODEBPARAM: ProcessSwitch=UE_EUTRAN_CAPB_ENQUIRY_OPT_SW-1;
//Enabling UE random access control based on cell radius
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=ACCESS_BYND_CELL_RAD_PROHIB_SW-1;
//Enabling optimized measurement of RRC connection setup success rate based on cell radius
MOD NRCELLFEATURESWITCH: NrCellId=0, CounterConfigSwitch=RRC_SETUP_SUCC_RADIUS_OPT_SW-1;
```

High-frequency TDD command examples

```
//Setting the percentage by which the PRACH preamble detection threshold increases (0% as an example)
MOD NRDUCELLPRACH: NrDuCellId=0, PrachThldIncreaseRate=0;
//Enabling narrow-beam probing
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, FullChannelCovEnhSwitch=NARROW_BEAM_PROBE_SW-1;
//Setting the threshold for triggering narrow-beam probing when narrow-beam probing is enabled (-108 as an example)
MOD NRDUCELLPRACH: NrDuCellId=0, NarrowBeamProbeThld=-108;
//Configuring the maximum number of preamble transmissions (for example, when narrow-beam probing and Flexible Dense Beam are both enabled)
MOD NRDUCELLPRACH: NrDuCellId=0, MaxPreambleTransCnt=N20;
//Enabling PRACH coverage enhancement
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch=PRACH_COV_ENH_SW-1;
//Enabling non-contention-based RA optimization
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch=CONTENTION_FREE_RA_OPT_SW-1;
//Enabling NSA CBRA statistics optimization
MOD GNBOAMPARAM: StatisticsStrategySwitch=NSA_CBRA_STATISTICS_OPT_SW-1;
//Enabling UE E-UTRAN capability query optimization
MOD GNODEBPARAM: ProcessSwitch=UE_EUTRAN_CAPB_ENQUIRY_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Low-frequency TDD command examples

```
//Disabling PRACH false alarm detection optimization
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch= PRACH_FALSE_ALARM_DET_ALGO_SW-0;
//Disabling PRACH capability optimization
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=PRACH_ACCESS_CAPB_OPT_SW-0;
//Disabling non-contention-based RA optimization
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch=CONTENTION_FREE_RA_OPT_SW-0;
//Disabling backoff control
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch=BACKOFF_SW-0;
//Disabling PRACH fixed delay compensation (using a slot configuration of 4:1 along with SS6 as an example)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, FrequencyBand=N41,
SlotAssignment=4_1_DDSU, SlotStructure=SS6;
MOD NRDUCELLTRPALGO: NrDuCellTrpld=0, FixedDelayCompensation=DELAY_0_4;
//Disabling NSA CBRA statistics optimization
MOD GNBOAMPARAM: StatisticsStrategySwitch=NSA_CBRA_STATISTICS_OPT_SW-0;
//Disabling UE E-UTRAN capability query optimization
MOD GNODEBPARAM: ProcessSwitch=UE_EUTRAN_CAPB_ENQUIRY_OPT_SW-0;
//Disabling optimized measurement of RRC connection setup success rate based on cell radius
MOD NRCELLFEATURESWITCH: NrCellId=0, CounterConfigSwitch=RRC_SETUP_SUCC_RADIUS_OPT_SW-0;
//Disabling UE random access control based on cell radius
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=ACCESS_BYND_CELL_RAD_PROHIB_SW-0;
```

High-frequency TDD command examples

```
//Disabling narrow-beam probing
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, FullChannelCovEnhSwitch=NARROW_BEAM_PROBE_SW-0;
//Disabling PRACH coverage enhancement
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch=PRACH_COV_ENH_SW-0;
//Disabling non-contention-based RA optimization
MOD NRDUCELLPRACH: NrDuCellId=0, RachAlgoOptSwitch=CONTENTION_FREE_RA_OPT_SW-0;
//Disabling NSA CBRA statistics optimization
MOD GNBOAMPARAM: StatisticsStrategySwitch=NSA_CBRA_STATISTICS_OPT_SW-0;
//Disabling UE E-UTRAN capability query optimization
MOD GNODEBPARAM: ProcessSwitch=UE_EUTRAN_CAPB_ENQUIRY_OPT_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Optimized counter measurement for contention-based RA: If the values of the **N.RA.Contention.Att** and **N.RA.Contention.Resp** counters decrease after this function is enabled, this function has taken effect.

Statistics collection optimization for contention-based RA in NSA scenarios: If the values of the **N.RA.Contention.Att** and **N.RA.Contention.Resp** counters decrease after this function is enabled, this function has taken effect.

PRACH false alarm detection optimization: If the value of **N.RA.Contention.Att** changes after this function is enabled with the **NRDUCellPrach.PrachFalseAlmDetDisThld** and **NRDUCellPrach.PrachThldIncreaseRate** parameters reconfigured, this function has taken effect.

PRACH fixed delay compensation: After this function is enabled, you can observe the number of times the TA values in RA response messages falls into different ranges (**N.RA.TA.UE.Index0** to **N.RA.TA.UE.Index15**) to check whether the TA values of accessing UEs correspond to the coverage distance after fixed delay compensation. If yes, this function has taken effect.

Activation verification is not involved in other sub-functions.

5.4.3 Network Monitoring

Observe the following counters to monitor RA:

- Counters related to contention-based RA: **N.RA.Contention.Att**, **N.RA.Contention.Att.Max**, **N.RA.Contention.Resp**, **N.RA.Contention.Msg3**, and **N.RA.Contention.Resolution.Succ**
- Counters related to non-contention-based RA: **N.RA.Dedicated.Att**, **N.RA.Dedicated.Resp**, and **N.RA.Dedicated.Msg3**

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification"
- *Uplink Scheduling*
- *5G Networking and Signaling*
- *mmWave Beam Management (High-Frequency TDD)*
- *High Speed Mobility*
- *Hyper Cell*
- *Cell Management*
- *Cell Combination*
- *UE Power Saving*
- *Carrier Aggregation*
- *Scalable Bandwidth*
- *Channel Management*
- *URLLC*
- *FWA*
- *MIMO (TDD)*
- *Standards Compliance*
- *RedCap*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Intelligent Aggregate Scheduling (High-Frequency TDD)*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *SON Management in MAE Product Documentation*